

SUSTAINABILITY & CLIMATE PORTFOLIO ↙

Presented by

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Shooting
better
everyday
experiences

sodexo | 

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Sodexo is committed to the practice of Integrated Reporting, based on the recommendations of the International Integrated Reporting Council (IIRC) and the Group's roadmap for corporate responsibility, Better Tomorrow. Managers from various departments within the Group took part in a series of workshops to cocreate the report, ensuring there is a common perspective on Sodexo's overall economic, social and environmental performance. This Fiscal 2025 Integrated Report draws on information from the Universal Registration Document in which it is published.



This is a translation into English of the (universal) registration document of the Company issued in French and it is available on the website of the Issuer.

The Universal Registration Document was filed on October 31, 2025 with the AMF, as competent authority under Regulation (EU) 2017/1129, without prior approval pursuant to Article 9 of the said regulation.

The Universal Registration Document may be used for the purposes of an offer to the public of securities or admission of securities to trading on a regulated market if completed by a securities note and, if applicable, a summary and any amendments to the Universal Registration Document. The whole is approved by the AMF in accordance with Regulation (EU) 2017/1129.

This Document is a reproduction of the official version of the Universal Registration Document including the 2025 Annual Financial Report prepared in accordance with the European Single Electronic Format (ESEF) and filed with the AMF, available on Sodexo's website, www.sodexo.com, and on the AMF website, www.amf-france.org.



Be the leader in Food & Services, shaping better everyday experiences at every moment in life

Key figures
as of August 31, 2025

24.1 billion euros

consolidated revenues

426,464

employees

27,000

client sites

#2

France-based private employer worldwide*

43

countries

80 million

consumers served daily

7.6 billion euros

in market capitalization

80%

employee engagement rate

Listed on the **CAC Next 20**,
CAC SBT 1.5, **FTSE 4 Good**
and **DJSI**.

Founded in Marseille in 1966 by Pierre Bellon, Sodexo has developed a unique responsible business model and an integrated service offering that creates value for all its stakeholders.

Since its creation, Sodexo's founding mission has positioned the company as a pioneer to meet the challenges of everyday life.

The Group stands out for its independence, its founding family shareholding and its business model. Through its activities, it strives to be the leader in Food and Services and shaping better everyday experiences at every moment in life.

Source Sodexo
* 2025 Forbes Global 2000 ranking.



2.2.2.1 Climate change (E1)

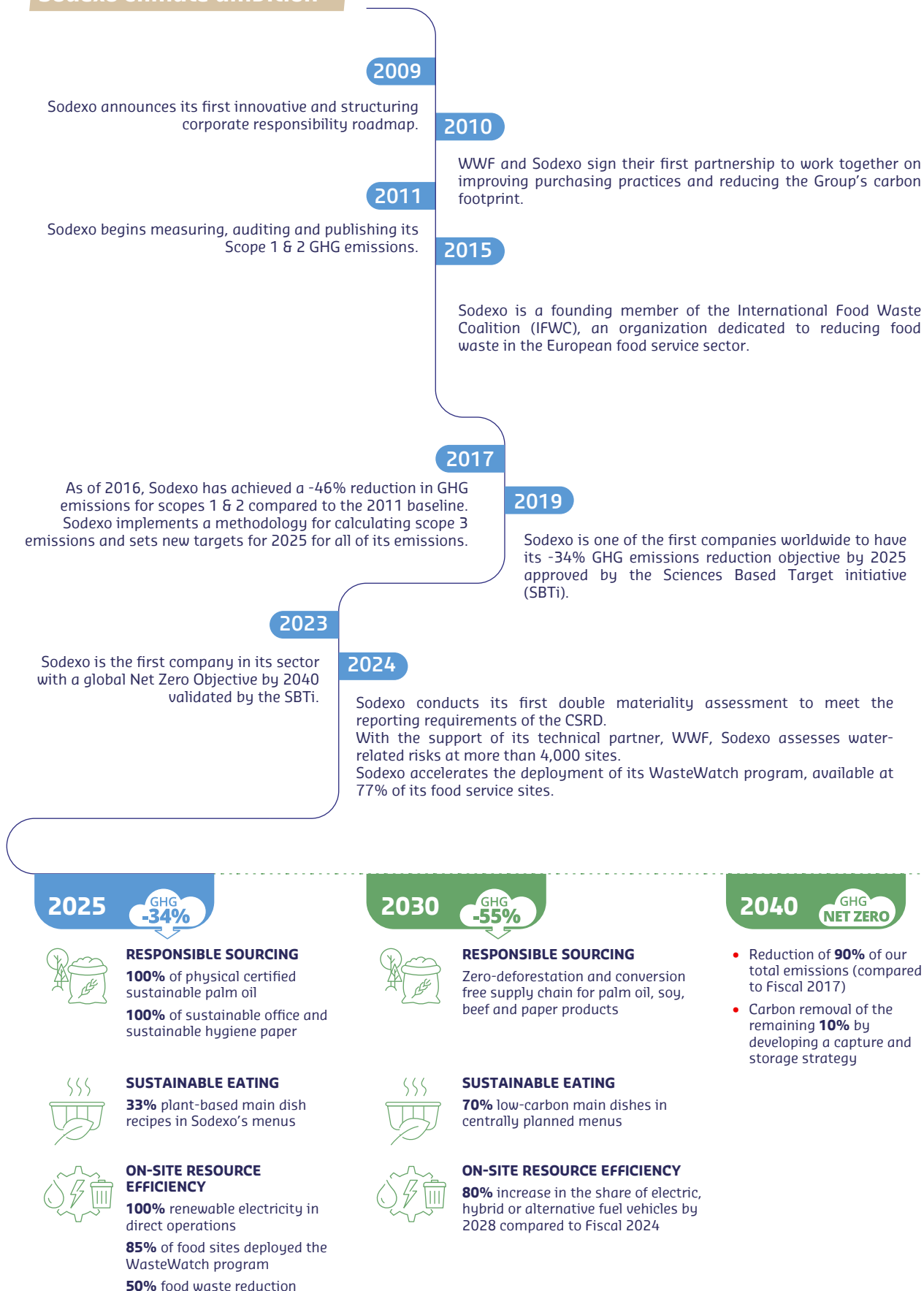
Impacts, Risks and Opportunities (IROs)

IRO Type	 Positive impact	 Negative impact	 Opportunity	 Risk
Time horizon	 Short-term	 Medium-term		
Value chain	 Upstream value chain	 Own operations	 Downstream value chain	

IRO type & horizon	IRO Name	Value chain	Sodexo's answer to these matters
E1 Climate change			
Resilience and adaptation to climate change			
 	Direct threat to the physical and psychosocial integrity of Sodexo employees in the event of a natural disaster caused by failure to adapt to climate change.		Sodexo faces both physical and transition risks linked to climate change, ranging from threats to the health, safety, and psychosocial well-being of employees and value chain workers during natural disasters, to the volatility of raw material prices caused by lower crop yields or geopolitical instability. To mitigate these impacts, Sodexo strengthens its health and safety governance, implements emergency preparedness in case of extreme weather events and business continuity plans, and supports employees and partners with appropriate resources. On the operational side, Sodexo invests in resilient infrastructure, renewable electricity, and energy-efficient technologies, balancing near-term capital expenditure with long-term cost savings and business resilience. By advancing responsible sourcing, supplier diversification, and regenerative agriculture practices, the Group reduces exposure to raw material price fluctuations while fostering more sustainable supply chains. Finally, through science-based targets, robust risk mapping, and CSRD-aligned reporting, Sodexo ensures compliance with evolving regulations, limits reputational and legal risks, and reinforces trust among clients, consumers, and stakeholders.
 	Direct threat to the physical and psychosocial integrity of Sodexo's value chain workers in the event of a natural disaster caused by failure to adapt to climate change.	 	
 	Volatility of raw material prices due to lower crop yields caused by climate change or geopolitical instability.		
 	Transition risk due to capital expenditure into infrastructures adaptation to climate change and higher operating costs for owned & leased assets.		
 	Business disruption and revenue loss due to significant destruction of assets or goods, caused by natural disasters or extreme weather events.	  	
 	Reputational and legal risk in the event of failure of equipment and services provided by Sodexo due to inappropriate adaptation to climate change.		
GHG emissions and energy			
 	Local green energy from biomass and solar energy production along the value chain.		Sodexo is promoting renewable energy use across its own operations, with a commitment to 100% renewable electricity by 2025 and local solar and biomass initiatives, which was reached as of August 2025. To reduce value chain emissions, which represent over 90% of its carbon footprint, Sodexo works with suppliers on science-based targets, responsible sourcing, and low-carbon meals. Robust governance and SBTi-validated targets ensure compliance with evolving regulations and mitigate reputational or legal risks. Finally, by embedding sustainability KPIs into client contracts and rolling out carbon labeling, Sodexo responds to growing client and consumer expectations for low-carbon services.
 	Contribution to climate change due to GHG emissions from the value chain.	 	
 	Reputational and legal risk in case of Sodexo's non-compliance with increasing local regulation and/or failure to achieve its GHG emissions related targets.		
 	Risk of market loss if failing to meet growing client and consumer expectations to deliver low-carbon services and in case of partnerships with high-emitting companies.		

As the first food services company to commit to achieving Net Zero emissions globally by 2040, we are accelerating our sustainability journey and transforming our operations to lead the industry toward a low-carbon future.

Sodexo climate ambition



To define this Net Zero strategy, built on a comprehensive climate transition plan, Sodexo initiated, in Fiscal 2021, a structured approach based on a standardized global climate risk assessment methodology to enable the identification and prioritization of material climate-related risks and opportunities.

Following this analysis, supported by scenario modelling designed to mitigate challenges and leverage opportunities across our operations and value chain, Sodexo developed its Net Zero strategy. This strategy is fully aligned with EU regulations and the Paris Agreement's 1.5°C goals. Our transition plan integrates robust governance, operational transformation, stakeholder engagement, and comprehensive risk management, ensuring a CSRD-compliant approach.

The plan has been formally reviewed and approved by the Board of Directors and members of the Sodexo Leadership Team. This demonstrates that climate action is overseen at the highest level of governance. The Group strategy team was an active stakeholder during the creation of the Climate Transition Plan, which ensured full

alignment with the company's strategic ambition and the acceleration of the Food services transformation.

It sets out a clear, time-bound decarbonization pathway, underpinned by interim and long-term targets for 2030 and 2040, along with a detailed implementation roadmap. To ensure its robustness and effectiveness, dedicated financial resources have been allocated to this work.

Material sustainability matters related to ESRS E1 Climate change are :

- Resilience and adaptation to climate change ;
- GHG emissions and energy.

Policies adopted to manage those material sustainability matters, corresponding actions, resources and targets are fully embedded in Sodexo's Climate transition plan and the 3 strategic core pillars that sustain Sodexo Climate & Nature strategy, described in section 2.2.2 Environmental information and summarized as follows :

Sodexo climate transition plan in order achieve Net Zero by 2040

Climate & Nature core pillars



Responsible sourcing

Policies

Sodexo commitment to protect forests and peatlands

Targets and commitments

- Deforestation and conversion free supply chain globally by 2030 for palm oil, soy, beef and paper products
- Encouraging and supporting suppliers to set Science-Based Targets aligned with the 1.5°C trajectory



Sustainable eating

Policies

Sodexo sustainability position paper - Low carbon meals

Targets and commitments

- 70% low-carbon main dishes in centrally planned menus by 2030
- 33% plant-based main dish recipes in its menus by 2025



On-site resource efficiency

Policies

No specific policy from this pillar is linked to ESRS E1

Targets and commitments

- 100% renewable electricity in its direct operations by 2025 and beyond
- 80% increase in the share of electric, hybrid or alternative fuel vehicles by 2028 compared to Fiscal 2024
- Food waste reduction: 85% deployment of the WasteWatch program by 2025 -50% reduction of food waste by 2025
- Implement energy efficiency actions, train teams and support client's transition to renewable electricity

Transition plan for climate change mitigation [E1-1]

NAVIGATING CLIMATE CHANGE RISKS AND OPPORTUNITIES [ESRS 2 IRO-1]

In Fiscal 2022, in order to identify and prioritize material climate-related risks and opportunities, we followed a three-step methodology, combining a hybrid approach that integrates "bottom-up" insights from operational teams with "top-down" oversight from senior management. The outcomes of this risk analysis also informed the more recent double materiality assessment conducted in Fiscal 2024.

1. Risk identification: the first step is the identification of risks that may impact Sodexo's ability to achieve its objectives, whether at site, country, regional or Group level. In addition to individual interviews with key stakeholders, we used other risk identification methods such as surveys and risk registers.

2. Risk evaluation: risks identified in the previous step are then evaluated using three risk criteria:

- (a) impact – the effect or consequence the risk will have,
- (b) likelihood – the frequency or probability of the risk occurring,
- (c) level of control – the level of control already in place to reduce the risk.

3. Risk prioritization: a cross-functional team in collaboration with external experts evaluated and prioritized climate-related risks and opportunities using the Task Force on Climate-Related Financial Disclosures (TCFD) framework, which categorizes risks into physical and transition risks:

(a) Physical risks: acute event-driven risks resulting from climate change such as an increase in severity in floods, cyclones or hurricanes, and chronic risks that result from shifts in longer term weather patterns such as higher temperatures causing more heatwaves;

(b) Transition risks: the policy, legal, technology and market change risks that may arise for a company transitioning to a lower carbon economy. Further reputational risk may also occur if a company fails to transition fast enough or meet stakeholder expectations in relation to climate change.

The assessment covers all phases of the value chain, as all types of risks are identified, for example difficulties to access resources (upstream), reputational risks related to clients and consumers (downstream), technology risks (direct operations, upstream and downstream).

26 physical and transition risks have been identified, all of which were considered relevant for Sodexo activities: policy and legal risks, technology-related transition risks, reputational risks, as well as chronic and acute physical risks. The magnitude of each risk occurring was assessed and broken down for Food and Facilities Management services. While likelihood or the probability of occurrence was considered during the evaluation phase, it is not disclosed to maintain focus on impact and relevance across Sodexo's services. In addition, the analysis was conducted using regional-level climate and economic data. While not yet based on geospatial coordinates for individual sites, the modelling reflects the diversity of Sodexo's operating environments and provides a robust foundation for assessing both physical and transition risks. The outcome of this analysis is detailed in the table below, including the different time horizons taken into account when assessing the impact.

During the risk assessment, we did not identify assets and business activities that are incompatible with or need significant efforts to be compatible with a transition to a climate-neutral economy.

Time horizon	 Short-term (1-5 years)	 Medium-term (5-10 years)	 Long-term (10+ years)
Value chain	 Upstream value chain	 Own operations	 Downstream value chain
Risk	1 Low	2 Medium	3 Significant 4 High

Description		Time horizon	Value Chain	Food Services	FM services
Physical risks					
Chronic					
Coastal erosion	Coastal erosion, driven by sea level rise and stronger storms, threatens properties and assets, causing damage to buildings and infrastructure.		 	3	2
Flooding – regulation	Regulations (including additional disclosure requirements, land use restrictions and new building standards) in response to increased flood risks.			2	1
Pests and diseases	Rising temperatures and shifting rainfall may trigger more crop pests and diseases, such as fungal blight or insect infestations, disrupting agricultural supply chains and raising food costs.		  	4	2
Reduced biodiversity	Reduced biodiversity from large-scale climate change, limiting access to natural resources.			4	1
Water use rights and regulations	Water scarcity leading to higher costs, stricter use regulations, and competition for water rights, including fishing restrictions.		 	4	4

Description		Time horizon	Value Chain	Food Services	FM services
Acute					
Hailstorms	More frequent and severe hailstorms from climate change causing damage to crops, buildings, equipment, property and infrastructure.			3	1
Land mass movements	More frequent landslides, sinkholes, and other land movements from climate change causing damage to buildings, equipment, property and infrastructure.			3	2
Windstorms	More frequent and intense windstorms (hurricanes, cyclones, typhoons, tornadoes) causing damage to crops, buildings, equipment, property and infrastructure.			3	1
Wildfires	More frequent and severe wildfires causing damage to buildings, equipment, property and infrastructure.			3	2
Chronic and Acute					
Drought	Prolonged droughts increasing water scarcity and reducing agricultural productivity			4	3
Flooding - damage	Flooding from rising sea levels, storms, and snowmelt, causing damages to crops, buildings, equipment, property and infrastructure as well as disrupting power supply.			4	2
Soil degradation	Soil degradation from heavy rainfall, shifting precipitation, and climate-driven biomass changes, reducing agricultural products quality and availability.			4	1
Temperature & humidity	More frequent and severe extremes in temperature and humidity.			4	2
Integrated Risks					
Ocean	Rising ocean temperatures and changes in acidity from climate change reducing seafood quality and availability.			3	1
Transportation - infrastructure	Sea level rise, erosion, and landslides damaging transport infrastructure, limiting access and service delivery.			3	3
Transition risks					
Policy and Legal					
GHG emissions regulations (including removal policies)	Stricter GHG policies (carbon tax, cap-and-trade, removal rules) requiring GHG emissions reductions and carbon capture.			4	4
Global trade policies	Stricter trade policies and regulations could place additional pressure on suppliers (including SMEs) to meet higher environmental standards, potentially leading to supply chain disruptions, reduced supplier capacity, or increased procurement costs for Sodexo.			2	2
Reforestation/afforestation	Government policies promoting reforestation and afforestation, leading to increased land competition, higher costs and fewer new buildings.			2	2
Removal of energy subsidies	Insufficient public support or reduced subsidies for renewable energy could slow the energy transition, limit access to affordable green electricity, and increase Sodexo's operational costs in some markets.			2	2
Transportation emissions reductions regulations	Tighter transport emissions regulations could raise logistics costs and operational complexity, especially where low-carbon infrastructure remains limited.			3	2
Technology					
Increased competition from low carbon and/or energy efficient technology	Growing competitiveness of renewable and energy-efficient technologies through policy, investment, and innovation.			3	3
Reputation					
Brand reputation	Reputational risks from consumer backlash over poor environmental practices or ties to unsustainable companies.			4	2
Both transition and physical risks					
Climate-induced social conflict & migration	Climate change driving inequality, migration, and increased risk of social conflict.			3	3
Consumer demand shift	Consumer demand shifting toward sustainable products and services.			4	2
Investor pressure	Rising investor pressure to improve environmental practices and cut GHG emissions.			2	2
Other land-use policies	Climate-driven land-use policies requiring sustainable farming, raising costs and reducing differentiation.			3	1

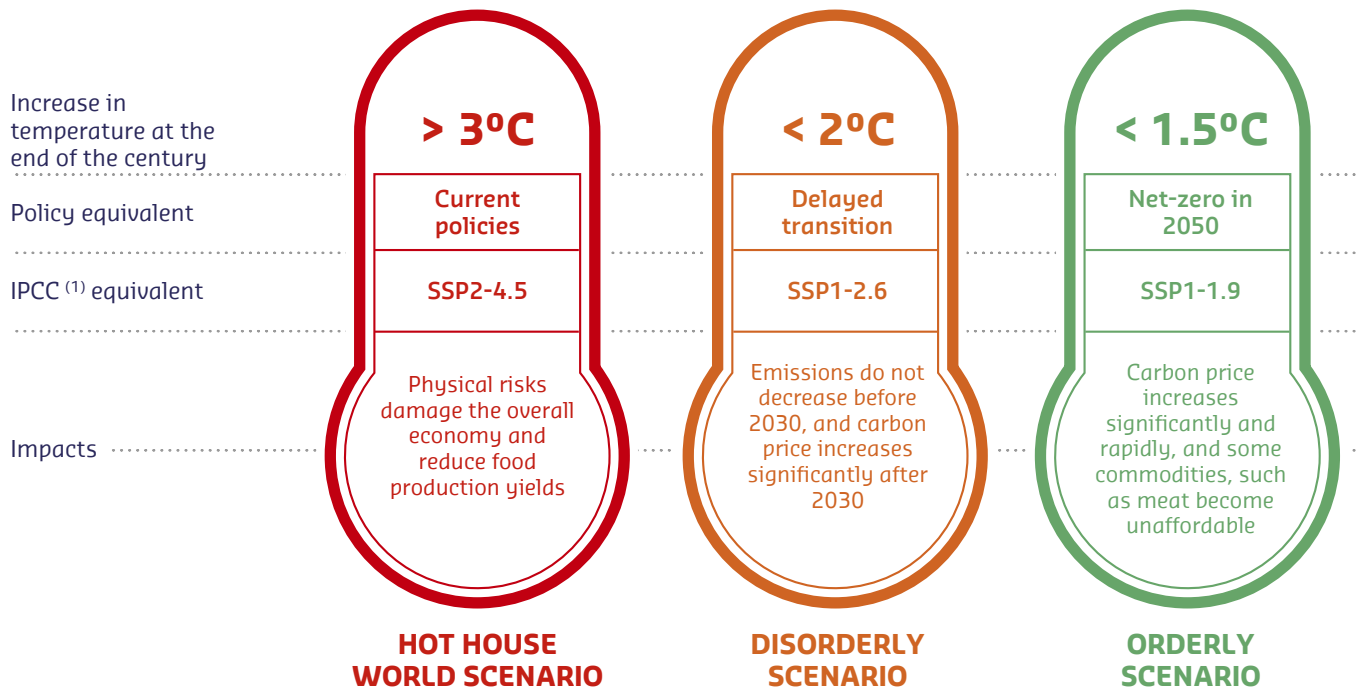
SCENARIO ANALYSIS [ESRS 2 SBM-3, E1-1]

The outcomes of the climate-related risk assessment helped Sodexo to proceed to a resilience analysis based on a scenario analysis, which is a core component of its climate transition plan.

The scope of the analysis, aligned with the TCFD methodology, covers both Food and Facilities Management services as well as upstream and downstream activities across the full value chain. The scenario analysis explores how Sodexo's business model may

respond under a range of plausible climate futures over the short- (to 2025), medium-(to 2030), and long-term (to 2040 and beyond).

Three main scenarios were identified and studied from the higher-emissions pathway to the one aligned with limiting global warming to 1.5°C with no or limited overshoot: Hot House World, Disorderly, and Orderly.



⁽¹⁾ Intergovernmental Panel on Climate Change.

Based on this analysis, Sodexo chose to align its climate transition plan with the orderly scenario, which is scientifically credible, consistent with the Paris Agreement, and most relevant to the Group's business model, global footprint, and stakeholder expectations. This scenario incorporates critical drivers such as evolving regulation (e.g. carbon pricing, deforestation-free supply chains), macroeconomic trends (e.g. food and energy price pressures), energy system transitions (e.g. electrification, renewable adoption), and technological developments (e.g. low-carbon kitchen equipment, digital emissions tracking). These factors directly influence Sodexo's procurement strategies, service design, and long-term investment decisions.

The climate scenarios used are fully compatible with the critical assumptions underlying Sodexo's financial statements. As outlined in section 2.2.2 of the consolidated financial statements, the potential long-term effects of climate-related risks, including those associated with supply chain disruptions, resource costs, and margin pressures, were integrated into the Group's strategic planning, risk management processes, and impairment testing assumptions.

Extensive internal modelling and sensitivity analyses confirmed that Sodexo's strategy is resilient under a 1.5 °C pathway, even though this scenario is the most ambitious and challenging to achieve. In particular, the modelling validated that expected changes in carbon pricing, demand for low-carbon services, shifts in dietary patterns, and required investments, such as those in carbon capture and storage post-2030, do not necessitate changes to asset useful lives, impairment assumptions, or carrying values at this stage.

This exercise also confirmed that the Group's long-term financial plans remain robust and aligned with its Net Zero 2040 target. While climate change introduces uncertainties, the scenario analysis demonstrates that Sodexo's strategic choices and operating model are compatible with a transition to a low-carbon economy and resilient under the most stringent global warming scenarios.

The findings from this analysis directly informed the development of Sodexo's Net Zero 2040 strategy and the operational levers supporting it, including menu redesign, energy optimization, supply chain transformation, and greener logistics aimed at both reducing emissions and enhancing the long-term resilience of the Group's business model.

Resilience and adaptation to climate change [ESRS 2 SBM-3, E1-1 E1-3, E1-4]

Building upon the scenario analysis and the selection of the Orderly Transition pathway, Sodexo has identified a set of concrete decarbonization levers that are central to achieving our Net Zero 2040 ambition. These levers are not only aligned with the macroeconomic, regulatory, and technological assumptions of the selected scenario, but also represent actionable initiatives already underway across our operations and value chain.

To achieve our Net Zero 2040 ambition and ensure the business resilience of the Company, Sodexo has committed to reducing its carbon footprint in the near- and long-term, in alignment with the SBTi guidance for the Forest, Land-Usage and Agriculture (FLAG) sector. All below targets are science-based, compatible with limiting global warming to 1.5°C. and were validated by the SBTi. They are presented in absolute reduction targets compared to a 2017 baseline and, for Scope 2 emissions, we are using market-based figures.

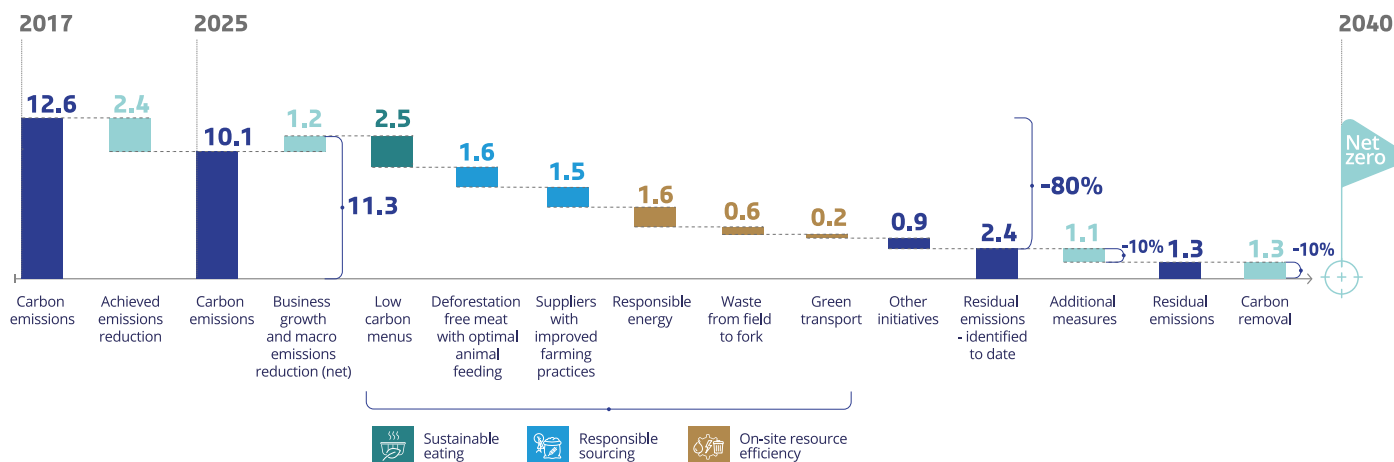
In calculating our carbon footprint, Sodexo accounts for all relevant greenhouse gases included in the CO₂e metric, specifically carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and selected fluorinated gases (HFCs, PFCs, SF₆, NF₃). These gases are converted into carbon dioxide equivalents using the latest 100-year Global Warming Potential (GWP100) values published by the Intergovernmental Panel on Climate Change (IPCC), ensuring methodological consistency and scientific rigor. This approach is applied comprehensively across Scope 1, Scope 2, and Scope 3 emissions. By adopting this robust accounting framework, Sodexo ensures that its emissions reporting is transparent, comparable, and aligned with the SBTi. This enables the company to track progress credibly and supports its commitment to achieving Net Zero greenhouse gas emissions by 2040.

Mid and long-term objectives:

- **by 2030, -55% total GHG emissions reduction**, of which:
 - -55% Scope 1&2 GHG emissions reduction;
 - no deforestation and conversion across primary deforestation-linked commodities;
 - -55% Scope 3 GHG emissions reduction, of which:
 - -40% Scope 3 Forest Land and Agriculture (FLAG) emission reduction.
- **by 2040, Net Zero**, of which:
 - -90% Scope 1&2 GHG emissions reduction;
 - -90% Scope 3 GHG emissions reduction, of which:
 - -72% Scope 3 FLAG emission reduction.

To address identified risks and seize the business opportunities related to climate change, the climate transition plan was structured around three core pillars: responsible sourcing, sustainable eating, and on-site resource efficiency. Each pillar is linked to specific actions and initiatives designed to reduce emissions across our value chain. These actions are not only climate-mitigation measures; they also represent strategic adaptations to ensure Sodexo's long-term resilience in a low-carbon economy. As Scope 3 emissions account for approximately 99% of Sodexo's total footprint, these decarbonization levers are primarily designed to tackle upstream and downstream emissions across the value chain, where the greatest impact and reduction potential lie. These three pillars are now fully embedded into Sodexo's strategic and operational planning as the primary levers for decarbonization and will enable Sodexo to reach its near- and long-term targets.

Sodexo Net Zero trajectory and decarbonization levers



To assess the effectiveness of these levers, we conducted a quantitative analysis of their carbon reduction potential. The results show that, if fully implemented, these initiatives can deliver over 80% of the emissions reductions required to meet our 2040 net-zero target. For the remaining 20%, we consider that:

- 10% reduction will be reached thanks to **additional measures** unidentified at this stage. Closing this gap will depend on collective action by governments, civil society, and the private sector to drive systemic changes in areas such as food production, dietary patterns, and energy infrastructure.
- remaining 10% will be reached thanks to **carbon removal**. In line with SBTi Net-Zero guidance, Sodexo intends to address residual emissions through the limited use of high-quality carbon removals generated beyond our value chain, whether natural or technological. The detailed strategy for these removals has not yet been finalized, as our current priority remains on emissions reduction and we are awaiting the updated SBTi guidance on removals.

CLIMATE TRANSITION PLAN IMPLEMENTATION [E1-1, E4-1]

To help its operational teams achieve the Group's carbon reduction objectives in 43 countries, Sodexo provides them with a low-carbon strategy analysis and planning solution named Carbon Trajectory Tool developed by its partner, Tennaxia. This tool offers a personalized action plan based on the identification of the main sources of carbon emissions and the efforts that can be made to reduce them. Via this solution, countries can choose from a catalog of more than 30 operational actions linked to each one of the climate strategy pillars. This enables countries teams to create carbon reduction plans, quantify their impact before launching their program and then measure the progress.

To support the implementation of the climate transition plan, additional mechanisms have been put in place:

- Leadership incentives aligned with Net Zero 2040: as detailed in section 7.3 Compensation, executive compensation is tied to progress on climate goals, including achieving 100% renewable electricity, reducing food waste and increasing the share of sustainable recipes, in line with Sodexo's science-based targets.
- Building internal climate expertise: Sodexo is investing in building internal climate expertise by upskilling employees across functions. Through targeted training on climate change, carbon accounting methodologies, and emission reduction levers, the company is fostering a workforce that is equipped to contribute meaningfully to its Net Zero 2040 ambition.

Allocation of financial resources for the climate transition plan [ESRS E1-1, E1-3]

Sodexo's climate transition plan is fully integrated into the Group's strategic and operational model. As an asset-light company, Sodexo does not operate industrial facilities or own significant physical infrastructure, which means its transition to a low-carbon business model does not rely on major capital expenditure programs. Instead, our decarbonization approach focuses on transforming the core of our value proposition: the way we source, design, and deliver food and services, which allows us to generate significant emissions reductions without heavy capital investment.

Approximately 70% of Sodexo's total GHG emissions are linked to purchased goods and services, particularly the food we buy and serve. As a result, most of our transition levers — and therefore our resource allocation — focus on responsible sourcing and sustainable eating. This involves shifting towards more plant-based and seasonal menus, influencing consumer choices through behavioral nudges and carbon labelling, and working closely with suppliers to improve agricultural practices, reduce embedded emissions, and eliminate deforestation from our supply chains. These initiatives do not require significant capital expenditure but rather strategic procurement choices and contractual partnerships aligned with our net-zero objectives.

Operational expenditure (OpEx) linked to the transition plan is primarily dedicated to building and strengthening internal capabilities and specialized expertise. This includes expanding our sustainability, sourcing, and menu innovation teams, as well as engaging external experts and consultancy partners to support our work on low-carbon menu design, supplier engagement, and science-based sourcing strategies. Additional OpEx relates to the deployment of digital tools and data systems that support emissions measurement, traceability, and performance tracking.

Over the medium to long term, Sodexo expects the financial resources allocated to the transition plan to remain stable and proportionate to business growth. The Group does not anticipate major changes to the useful life of assets, nor significant capital investments directly related to the decarbonization strategy. Instead, the reallocation of existing resources, combined with a stronger focus on procurement, innovation, and partnerships, will remain the primary driver of emissions reduction and support the Group's trajectory to Net Zero by 2040.

The Group's climate transition plan is also the result of a comprehensive analysis conducted in 2022, which included a dedicated assessment of its potential financial implications. As part of this exercise, Sodexo evaluated the expected impacts of the transition plan on both CapEx and OpEx. The results confirmed that, given the Group's asset-light model and the strong focus of the plan on business transformation rather than infrastructure investments, the financial impacts are expected to remain limited compared to the Group's overall expenditure. Since this initial analysis, investments have already been directed towards strengthening our internal sustainability capabilities — including the expansion of dedicated teams, training programs for employees on climate-related topics, and specific training for chefs on sustainable cooking practices. In addition, Sodexo has invested in a carbon trajectory tool, now available across all countries, to support the effective monitoring and implementation of our Net Zero strategy. As a result of these early actions, no significant additional OpEx or CapEx is anticipated in the coming years, and the financial resources dedicated to the climate transition plan are expected to remain stable and proportionate to the Group's activity.

GHG REMOVALS AND GHG MITIGATION PROJECTS FINANCED THROUGH CARBON CREDITS [E1-7]

Sodexo does not currently engage in GHG removals or finance GHG mitigation projects through carbon credits. At this stage, the Group is focused on reducing emissions across its own operations and value chain. In line with SBTi Net-Zero guidance, Sodexo recognizes that high-quality carbon removals may play a role in addressing residual emissions. We are therefore monitoring market developments and the forthcoming SBTi guidance to determine whether, and under what conditions, such instruments could be relevant for Sodexo in the future.

INTERNAL CARBON PRICING [E1-8]

Sodexo does not have a formal, internal carbon pricing mechanism in place. Over the past years, the Group has been using carbon prices for decision-making in some specific projects.

In 2022, as part of the scenario analysis, Sodexo has used the carbon price as a key variable impacting Sodexo, when estimating the impact on Sodexo Underlying Operating Profit (UOP) on short, medium and long term. This variable was calculated by considering regulation that may increase the price of carbon globally.

A carbon price aligned with the prices of regulated carbon taxes are also used occasionally, such as in some acquisition projects, or assessment of specific activities. In this case, it is used as a shadow price, to assess an approximate price of aligning those specific projects with Sodexo ambitions.

Outside of those specific instances, Sodexo does not use internal carbon prices in decision-making, as it does not apply to the way it operates services and projects.

ANTICIPATED FINANCIAL EFFECTS FROM MATERIAL PHYSICAL AND TRANSITION RISKS AND POTENTIAL CLIMATE-RELATED OPPORTUNITIES [E1-9]

Sodexo has not yet quantified or reported the anticipated financial effects of material physical and transition risks, nor the potential climate-related opportunities, in the current risk assessment. In line with CSRD requirements and evolving best practice, Sodexo will start the assessment of the anticipated financial effects to ensure that the financial implications of climate-related risks and opportunities are systematically considered.

Stakeholder & policy engagement

Driven by its desire to be a responsible partner to its clients and suppliers with common sustainability goals, Sodexo has been working collaboratively with a broad spectrum of stakeholders, including NGOs, government bodies, regulators, investors, and think tanks, not only to support the development and implementation of robust environmental standards, but also to actively contribute to climate-related policy advocacy.

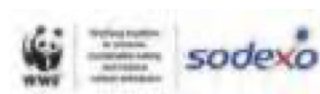
Policy advocacy

Sodexo supports strong climate action through the participation in trade associations and partnerships with NGOs and other association.

At country level, Sodexo also engages with local associations and regulators. Sodexo is a member of AFEP (*Association Française des Entreprises Privées*), the French association of private companies. Sustainability is now a central priority for AFEP's member companies: reduction of GHG emissions, acceleration of circular economy practices, control of industrial risks, and development of

practical experimental approaches in conjunction with the public authorities. Sodexo Chairwoman of the Board of Directors and Chief Executive Officer sits at the AFEP board.

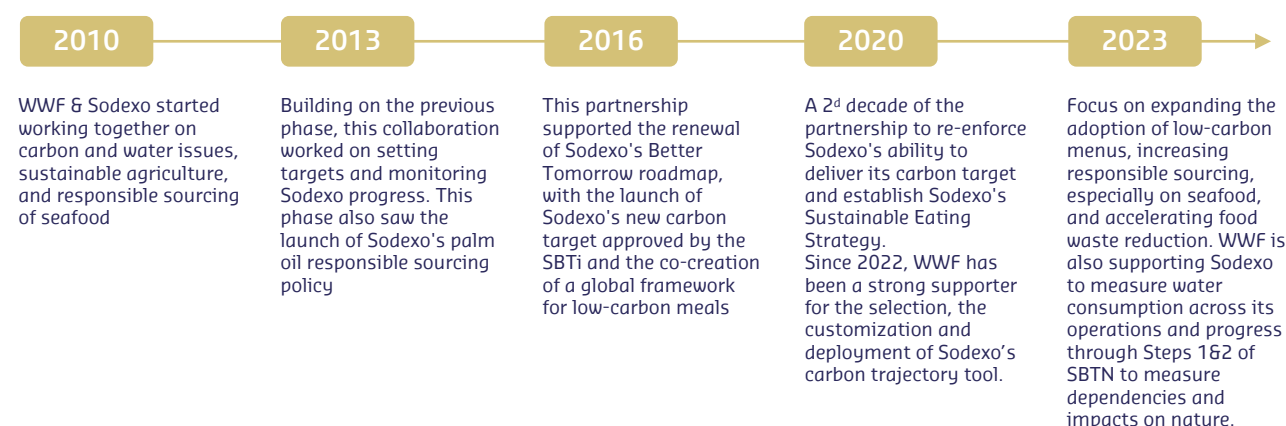
World Wildlife Fund for Nature (WWF) partnership



Sodexo has been a partner of WWF since 2010 to understand, measure and reduce its environmental footprint. Sodexo and WWF have carried out

projects to address the climate urgency since then on deforestation, sustainable eating, responsible sourcing, and food waste. The partnership brings WWF's technical expertise and thought leadership on sustainability to support further reducing the impact of Sodexo's operations on the environment.

Sodexo and WWF also advocate jointly for stronger government regulation on topics such as sustainable food systems, deforestation, sustainable fish and seafood and food waste.



Other multi-stakeholder initiatives

Sodexo believes global decision-makers must work together to help ensure the future well-being of the planet. It therefore participates in numerous initiatives such as the Champions 12.3 coalition, ReFED, WRAP, the Consumer Goods Forum, REFRESH, and Food Service Europe. The Group is also a founding member of the International Food Waste Coalition, which fights food waste by way of its collaborative "farm-to-plate" approach.

Sodexo Climate & Nature actions go beyond emissions. We are progressively identifying other significant climate-related impacts across our value chain, including land use changes, and biodiversity loss associated with food production. Our operations also depend on climate-sensitive resources such as agricultural commodities, water availability, and energy supply. These dependencies are mapped across our business to anticipate disruption risks.

In Fiscal 2023, Sodexo conducted a Water Footprint Indicator as well as a water risk assessment to evaluate our impacts and dependencies to the water resource and assess the exposure of client's sites to water risks. In Fiscal 2024, Sodexo also conducted a nature and biodiversity risk assessment using the Science Based Targets Network (SBTN) Steps 1&2 methodology. The SBTN assessment framework aims to identify priority locations, commodities and business units related to Water, Land Use, pollution and biodiversity. This assessment showed that 99% of Sodexo's nature and biodiversity impact occurs in its supply chain. The assessment confirmed Sodexo's supply chain focus in its biodiversity approach as well as its priority focus on deforestation and conversion and sustainable fish and seafood. The methodology and assessments results will allow Sodexo to further refine its strategy and targets in the coming years, these insights inform our sustainability roadmap and supplier engagement strategies.

Scope 1, 2 & 3 GHG emissions accounting

To ensure accountability and continuous improvement, Sodexo tracks progress against its climate targets through defined KPIs, including GHG emissions (absolute emission and intensity), share of renewable electricity in the direct site operations, low-carbon meals, etc. Performance is reviewed annually with course corrections made when needed. Detailed results on GHG emissions and energy are presented in section below. In addition, Sodexo ensures transparency and progress tracking through EU Taxonomy disclosures, mapping economic activities to sustainability criteria and CapEx/OpEx alignment (refer to section 2.2.5 Green Taxonomy).

GHG emissions footprint

For Sodexo, as a service provider operating primarily on client sites, the vast majority of greenhouse gas emissions fall within Scope 3. These emissions stem largely from purchased goods and services, particularly food, as well as upstream and downstream activities across our value chain. Addressing Scope 3 is therefore at the heart of our climate strategy, requiring close collaboration with suppliers, clients, and consumers to reduce impacts where they are most significant. By focusing on responsible sourcing, sustainable food offers, and operational efficiency, Sodexo tackles its largest emissions category while creating long-term value for stakeholders.

Scope 3 Indirect GHG emissions

Scope 3 includes indirect emissions from the value chain: upstream and downstream of Sodexo's activities generated by various stakeholders: suppliers, consumers, service providers.

Sodexo does not directly derive revenues from activities related to coal, oil, or high-intensity power generation, and none of our operations have been identified as causing significant harm to environmental objectives under the EU Taxonomy.

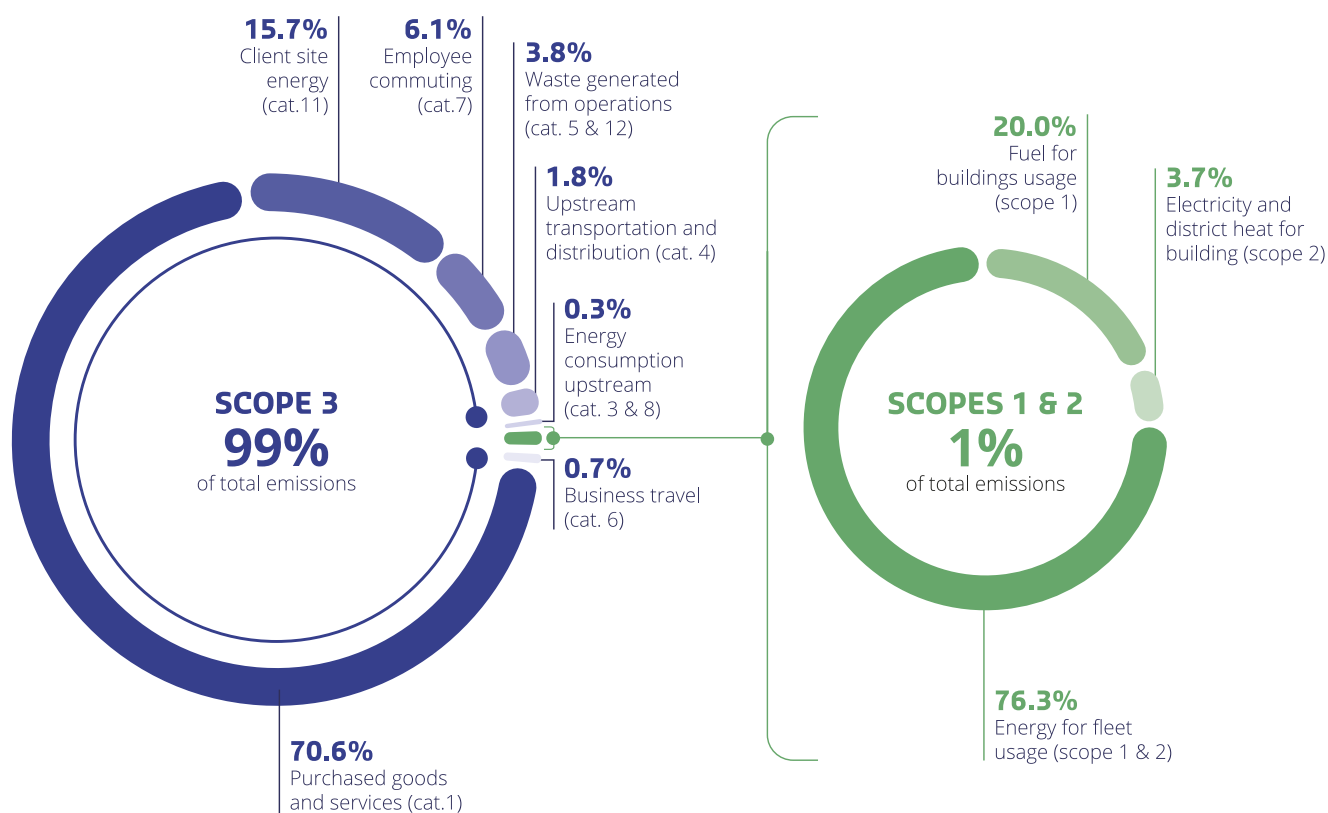
Sodexo monitors and reports its GHG emissions in accordance with the GHG Protocol as well as the progress towards its climate-related targets.

Locked-in GHG emissions mitigation

Sodexo has conducted a qualitative assessment of its potential locked-in GHG emissions. Given Sodexo's asset-light business model, the company does not identify significant locked-in emissions that could jeopardize its GHG reduction targets or pose a transition risk. The primary sources of locked-in emissions include right-of-use assets, such as vehicles and office buildings, which are structured under relatively short-term agreements and gives the flexibility to adapt swiftly to emerging low-carbon technologies and mitigation strategies. While acknowledging the potential impact of locked-in emissions, Sodexo remains confident that its Climate Transition Plan will not be hindered, as the company's operations do not rely on long-lived, carbon-intensive assets.

Scopes 1 & 2 Direct GHG emissions

Scopes 1 and 2 pertain to the company's direct emissions and indirect emissions linked to the production of the energy used by the company.



The methodological note (in section 2.2.6 Reporting methodology) includes the main principles, including re-baselining procedures where applicable, and limitations retained for the calculation of Sodexo GHG emissions.

GHG emissions and energy

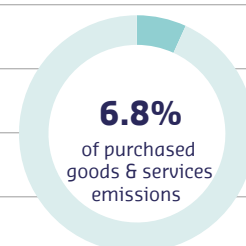
FOOD PRODUCTS VOLUMES AND THEIR GHG EMISSIONS IMPACT

Categories	Volume (% of kg)	Emissions (% of tCO ₂ e)
 Meat	8.5% 	 37.4%
 Ready made	5.8% 	 12.2%
 Dairy & eggs	12.4% 	 11.2%
 Baked goods	5.7% 	 10.0%
 Coffee, tea, cocoa & beverages	33.3% 	 7.8%
 Fruits & vegetables	21.9% 	 6.0%
 Fish & seafood	1.1% 	 3.8%
 Beans, pulses & cereals	6.2% 	 3.6%
Other commodities	N/A	 3.4%
 Sugar, spices & oils	2.7% 	 2.6%
 Sauce & canned food	2.5% 	 2.0%



NON FOOD PRODUCTS SPEND AND THEIR GHG EMISSIONS IMPACT

Categories	Volume (% of spend)	Emissions (% of tCO ₂ e)
Maintenance & services	69.1% 	 54.4%
Cleaning chemical & products	6.7% 	 17.4%
Uniforms & linen	5.0% 	 5.9%
Food services disposables	4.9% 	 5.2%
Reusable containers, dishes & cutlery	3.5% 	 4.3%
Hygiene paper, office paper and other paper products	5.7% 	 5.3%
Non food plastic	2.5% 	 3.2%
Other non-food products	0.7% 	 1.6%
Personal protection equipment (PPE)	2.0% 	 2.6%



Metrics

ENERGY CONSUMPTION AND MIX [E1-5]

(in Megawatt- hours)		FISCAL 2025
Fossil energy consumption		400,748
Share of fossil sources in total energy consumption (%)		75.2 %
Consumption from nuclear sources		302
Share of consumption from nuclear sources in total energy consumption (%)		0.1 %
Fuel consumption for renewable sources, including biomass (also comprising industrial and municipal waste of biologic origin, biogas, renewable hydrogen, etc.)		12,036
Consumption of purchased or acquired electricity, heat, steam, and cooling from renewable sources		119,979
Consumption of self-generated non-fuel renewable energy		0
Total renewable energy consumption		132,015
Share of renewable sources in total energy consumption (%)		24.8 %
TOTAL ENERGY CONSUMPTION		533,065

	FISCAL 2025	FISCAL 2024	CHANGE
% of electric, hybrid or alternative fuel vehicles in Sodexo fleet	29.8%	23.2%	+6.6 pts
% renewable electricity consumed in our direct operations	96.9%	73.0%	+23.9 pts

The total energy consumption of the group amounts to 533,065 MegaWatt- hours, of which 24.8% from renewable sources. This encompasses electricity and other fuel types used for direct operations (buildings and fleet).

The Group achieved 96.9% of renewable electricity in our direct operations, compared to 73.0% in the previous year, reaching its 2025 RE100 commitment in all countries except for specific RE100 allowed exclusions for Fiscal 2025 (for example, due to limited availability of renewable electricity and credits). In Fiscal 2025, 36 countries were sourcing 100% renewable electricity up from 15 countries in Fiscal 2024.

The Group makes use of Energy Attribute Certificates (EACs) as contractual instruments to substantiate the purchase of renewable electricity and reduce the market-based scope 2 emissions. These include, Guarantees of Origin (GOs), International Renewable Energy Certificates (I-RECs), Renewable Energy Certificates (RECs), which cover 22% of our electricity consumption for Fiscal 2025.

The percentage of electric, hybrid or alternative fuel vehicles in the Sodexo fleet represents 29.8% at the end of Fiscal 2025, compared 23.2% at the end of Fiscal 2024. Sodexo already achieved its 2026, 20% target and is on track to reach 35% by the end of Fiscal 2027.

GHG emissions disaggregated by Scopes 1 & 2 and significant Scope 3 categories [E1-6]

tCO ₂ e	FISCAL 2025	FISCAL 2024	CHANGE	BASELINE FISCAL 2017	REDUCTION COMPARED TO BASELINE
Scope 1 GHG emissions					
Gross Scope 1 GHG emissions	90,224	93,773	-3.8%	107,464	-16.0%
Percentage of Scope 1 GHG emissions from regulated emission trading schemes (%)	— %	— %	—%		—%
Scope 2 GHG emissions					
Gross location-based Scope 2 GHG emissions	22,814	23,588	-3.3%	44,727	-49.0%
Gross market-based Scope 2 GHG emissions	3,908	12,809	-69.5%	43,653	-91.0%
SCOPE 1 & 2 GHG EMISSIONS - MARKET-BASED	94,132	106,582	-11.7%	151,117	-37.7%
Significant scope 3 GHG emissions					
Purchased goods and services - FLAG	4,748,615	4,826,516	-1.6%	6,267,921	-24.2%
Purchased goods and services - Non-FLAG	2,417,103	2,514,496	-3.9%	2,644,449	-8.6%
1 - Purchased goods and services	7,165,718	7,341,012	-2.4%	8,912,370	-19.6%
3 - Fuel- and energy-related activities	27,411	28,598	-4.2%	32,020	-14.4%
4 - Upstream transportation and distribution	187,539	170,406	10.1%	445,619	-57.9%
5 - Waste generated in operations	61,115	66,907	-8.7%	82,358	-25.8%
6 - Business travel	69,164	79,567	-13.1%	40,267	71.8%
7 - Employee commuting	621,048	732,647	-15.2%	775,650	-19.9%
8 - Upstream leased assets	1,084	1,515	-28.4%		—%
11 - Use of sold products	1,593,324	1,610,595	-1.1%	1,678,815	-5.1%
12 - End-of-life treatment of sold products	322,557	328,149	-1.7%	454,400	-29.0%
Scope 3 GHG emissions	10,048,959	10,359,396	-3.0%	12,421,499	-19.1%
Scope 3 GHG emissions - FLAG	4,748,615	4,826,516	-1.6%	6,267,921	-24.2%
Scope 3 GHG emissions - Non-FLAG	5,300,344	5,532,880	-4.2%	6,153,578	-13.9%
TOTAL GHG EMISSIONS					
Total location-based GHG emissions	10,161,998	10,476,757	-3.0%	12,573,690	-19.2%
Total market-based GHG emissions	10,143,091	10,465,978	-3.1%	12,572,616	-19.3%
Market-based Scope 1 & 2 GHG emissions intensity, per revenue⁽¹⁾ (tCO₂e/ m€)	3.9	4.5	-13.5%	7.7	-49.2%
Scope 3 GHG emissions intensity, per revenue⁽¹⁾ (tCO₂e/ m€)	417.4	439.5	-5.0%	632.1	-34.0%
Location-based Total GHG emissions intensity, per revenue⁽¹⁾ (tCO₂e/ m€)	422.1	444.5	-5.0%	639.9	-34.0%
Market-based Total GHG emissions intensity, per revenue⁽¹⁾ (tCO₂e/ m€)	421.3	444.0	-5.1%	639.8	-34.1%

(1) Fiscal 2025 revenues as per consolidated financial statements under IFRS (refer to section 4.1.1 Consolidated income statement). Prior year intensity and baseline is calculated based on revenues after re-baselining, as described in section 2.2.6 Reporting methodology.

Emissions Overview

Scope 1 & 2 emissions represent just 1% of Sodexo's total footprint, mainly from fleet energy (76.3%) and building fuel (20%). Scope 3 accounts for 99%, driven by Purchased Goods & Services (70.6%), Client Site Energy (15.7%), and Employee Commuting (6.1%).

Within Purchased Goods & Services, food accounts for 93.2% of emissions. Animal proteins have a disproportionate impact: meat represents only 8.5% of food volume but 37.4% of food-related emissions.

Total Scope 1 & 2 and 3 Emissions Progress

Sodexo reduced total Scope 1, 2, and 3 emissions by -19.3% from the 2017 baseline and by -3.1% YoY. Emission intensity fell -34.1% since 2017 and -5.1% year-on-year.

Progress is strongest in Europe and North America (-33%), while reductions are more challenging in high-growth markets, which account for 36% of emissions.

Scope 1 & 2 Progress

Scope 1 & 2 emissions decreased -11.7% YoY and -37.7% since 2017, surpassing the -34% 2025 target. This was driven by a major rise in renewable electricity use — now 96.9%, up from 73.0% in 2024 — reaching the RE100 commitment in all countries except for specific RE100 allowed exclusions (for example, due to limited availability of renewable electricity).

In Fiscal 2025, biogenic CO₂e emissions from the combustion or bio-degradation of biomass (in particular CH₄ and N₂O) represented around 2,379 tons in relation to scope 1 emissions.

Scope 3 Progress

Scope 3 emission intensity is down -34% compared to 2017, reflecting strong progress through sustainable sourcing, plant-based menu innovation, and food waste reduction.



Protasius Isyudanto
Partner



Lidwina Pradipta Putri
Associate - Sustainability



K A L A
A D V I S O R Y

July 2026

Southeast Asia Climate Technology Spotlight :

*Issue 2 of Kala Advisory's periodic
Southeast Asia Business Sector Spotlight*

In a glance

For European and East Asian businesses, Southeast Asia's climate technology sector offers **a region rich in feedstock and natural carbon sink potential**, backed by fast-maturing carbon markets, yet **still missing the proven carbon removal and capture technology** that net-zero targets now demand.

Executive summary:

Southeast Asia's climate technology market is maturing, but its capital and its carbon potential sit in different places. Private green investment reached 8 billion US dollar in 2024, yet **it flows mostly to renewable power, while the region's vast natural carbon sinks and abundant biomass remain underused** and the carbon removal sector stays thinly funded. Eight ASEAN states now target net zero, which requires not only cutting emissions but removing carbon and the market infrastructure to reward removal is finally arriving, which includes the ASEAN Common Carbon Framework, national registries and Article 6 of Paris Agreement that let regional credits reach international buyers. **This gap is the opportunity.**

The region holds the raw material. What it lacks is proven technology, credible verification and patient capital. European firms bring certification and compliance credibility built on the EU ETS and CRCF, while East Asian firms bring deployment-ready technology and manufacturing proximity. **For entrants to the region in this sector, build the supply base in Indonesia where the material is largest and Article 6 trading is now enabled and route the output through Singapore as Asia-Pacific's carbon trading hub, and reach the net-zero buyers who already exist abroad.**

Why Southeast Asia, why now

Southeast Asia is among the world's most climate-exposed regions and its coal-heavy growth keeps emissions rising, yet it holds a major share of the world's natural carbon sinks. In 2026, the carbon-market infrastructure to turn that potential into revenue arrived. The ASEAN Common Carbon Framework is harmonising credit standards, national registries are launching and Article 6 of Paris Agreement are opening the door for Southeast Asian credits to reach international buyers. The region is shifting from holding carbon potential to being able to sell it and the technology to unlock it is more on demand than ever.

1 Southeast Asia

Climate Tech Sectoral Structure

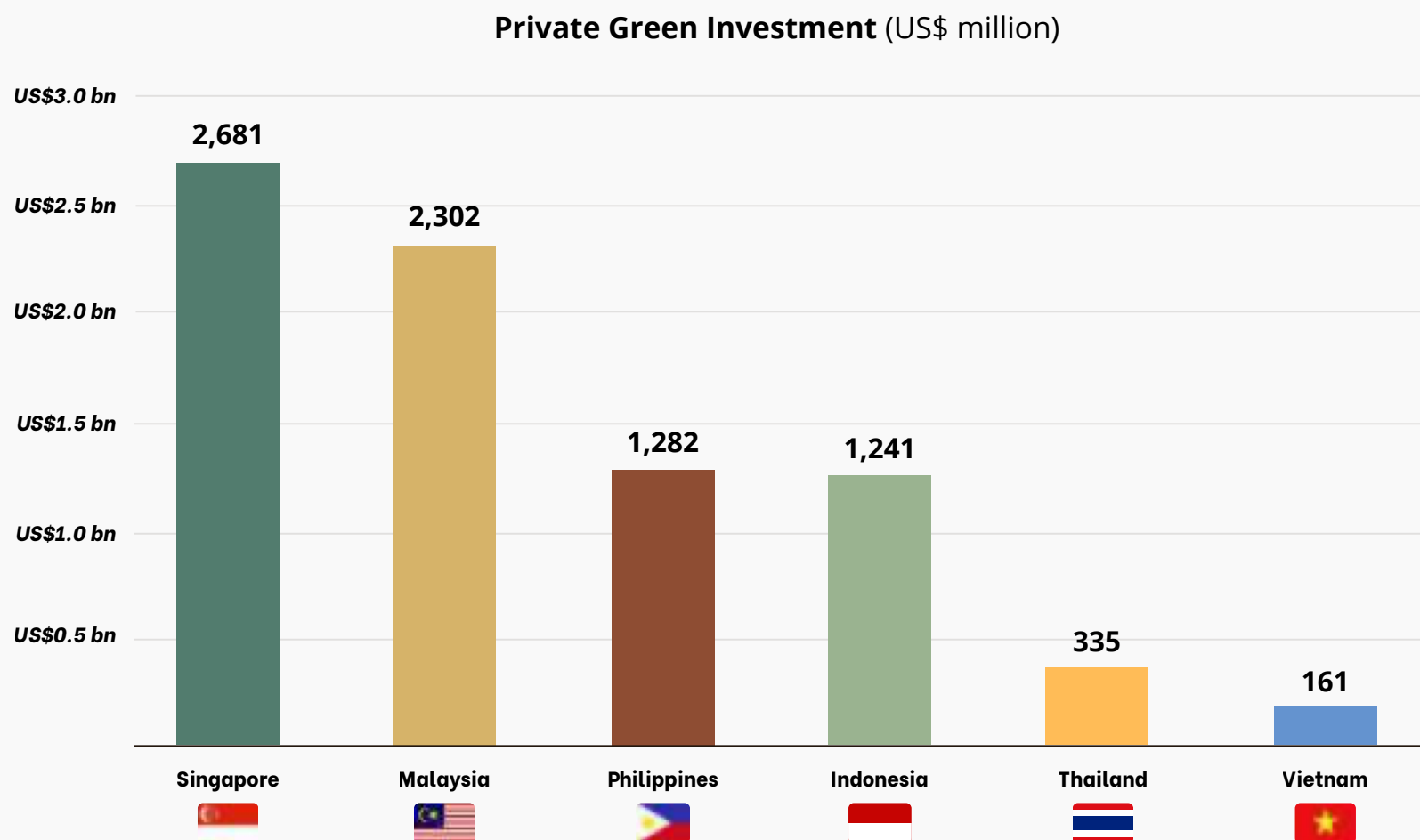


Figure 1: Private green investment in ASEAN-6 countries in 2024 (in US\$ million)

Southeast Asia's high climate vulnerability and emission-heavy growth make the region an ideal market for climate technology to cut and remove carbon. At the same time, the region holds a crucial piece of the global solution, which is its massive natural carbon sinks, a potential that climate technology can unlock through scalable nature-based solutions. Driven by emerging national carbon markets, expanding climate financing and regional net-zero commitments, private green investment in Southeast Asia reached US\$8 billion in 2024, representing a 43% year-on-year (yoy) increase from 2023.

However, comparing private green investments with nature-based carbon offset potential reveals a significant geographic disconnect. Capital is heavily concentrated in financial hubs like Singapore (accumulating US\$2.68 billion in capital), while the vast majority of carbon sink capacity sits in land-rich countries like Indonesia (201 megatonne CO₂e/year).

1 Southeast Asia

Climate Tech Sectoral Structure

Nature-based Solutions Carbon Offset Projected
Supply by 2030 in Southeast Asia
(Mt CO₂e/year)

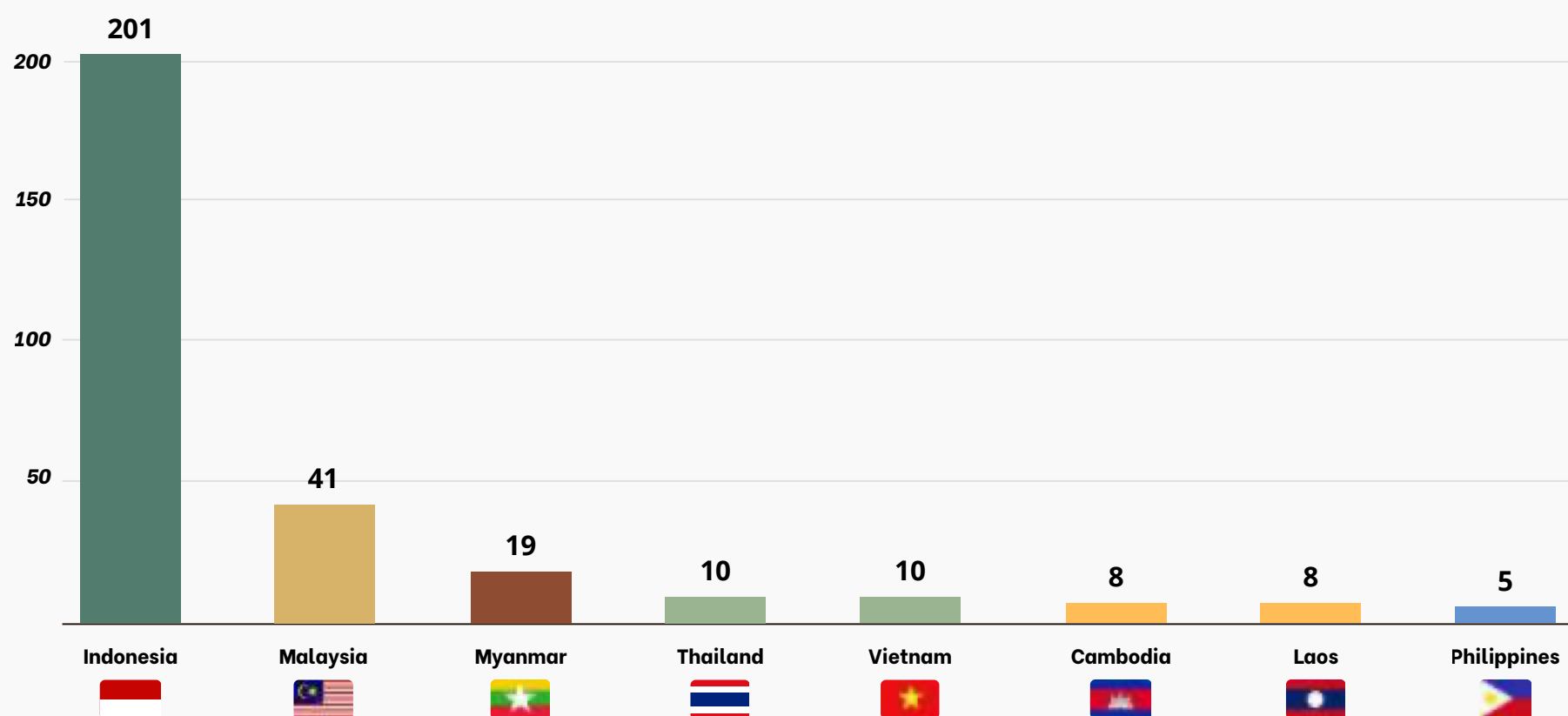


Figure 2: Nature-based solutions carbon offset projected supply by 2030 in Southeast Asia (in Mt CO₂e/year)

In 2024, eight countries in Southeast Asia target net-zero, which requires not only emission reduction but also carbon removal. Yet green capital overwhelmingly favours renewable power like solar, wind, and green hydrogen, which accounts for nearly two-thirds of total investment to prevent new emissions, while funding for climate projects in the agricultural and nature domains actually declined. This creates a striking paradox: **despite holding vast natural carbon sinks and abundant biomass feedstocks, the region's nature-based solutions (NbS) tech ecosystem remains in its nascent stages and underfunded.**

Implications for new entrants

Rather than competing in saturated power markets, experienced entrants can capture early market share by deploying mature technologies that unlock the region's vast nature-based feedstocks.

Recent catalysts

The [ASEAN Common Carbon Framework](#) is entering implementation, harmonising credit standards regionally, with national policies following (Indonesia's SRUK and Malaysia's carbon market policy). [Article 6 of Paris Agreements](#) bilateral agreements connect regional credits to international buyers, giving projects a legal route from local asset to revenue.

2 The European Angle

Europe brings established carbon-market experience to Southeast Asia, with its certification standards and regulations. The EU Emissions Trading System (ETS) has operated since 2005, and the Carbon Removal and Carbon Farming Framework (CRCF) was adopted in 2024, setting the standards for durable removal which is the kind of market infrastructure that Southeast Asia is now building. European development agencies have already worked across ASEAN on carbon rules and green finance. **That groundwork opens the door for European companies to bring the technology and problem-solving the region's carbon sector still lacks.**

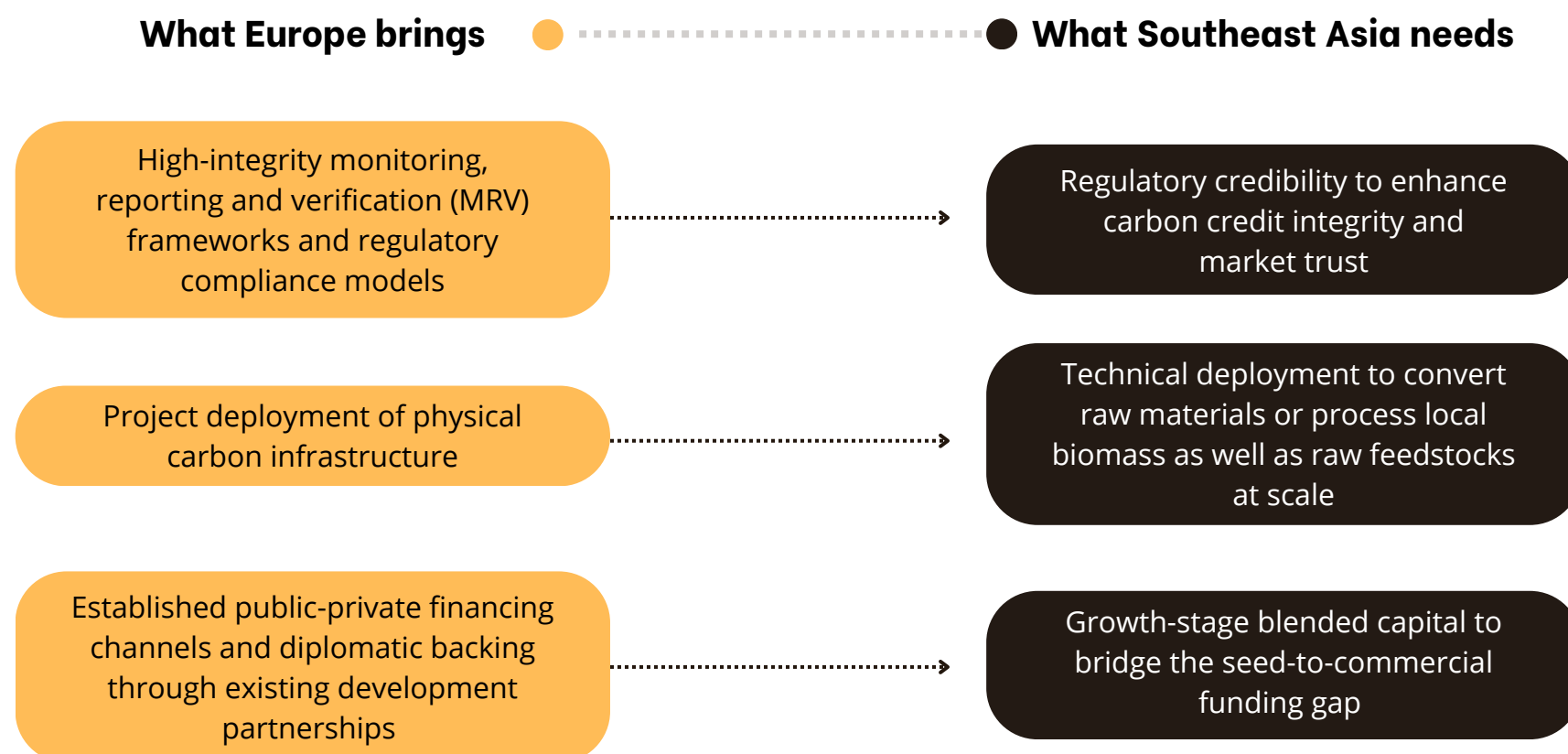


Figure 3: European capability mapping against Southeast Asia's needs

Implications for European entrants

Europe's strengths map directly onto the region's gaps: MRV and compliance credibility answer the trust deficit, project experience answers the deployment gap and access to blended finance bridges the funding gap local startups cannot cross. The European entrant's advantage is not the technology alone, it is arriving already certified, fundable and trusted by the institutions that shape the region's carbon rules.

3 The East Asian Angle

East Asia brings deployment-ready climate hardware and proximity to Southeast Asia.

Companies from China, Hong Kong, Japan and South Korea commercialise technology through engineering ecosystems the region has not yet built and proving it in larger markets first. East Asian public finance is also arriving at scale. For instance, the US\$221 million Green Climate Fund/Korean Development Bank Climate Technopreneurship Programme funds incubation and acceleration platforms to improve the local ecosystem for climate technology in Cambodia, Indonesia, Laos, Philippines and Vietnam. **For East Asian companies, Southeast Asia is a commercially promising region, with their long-standing economic cooperation, close proximity and cultural familiarity make entry a natural extension rather than a leap.**

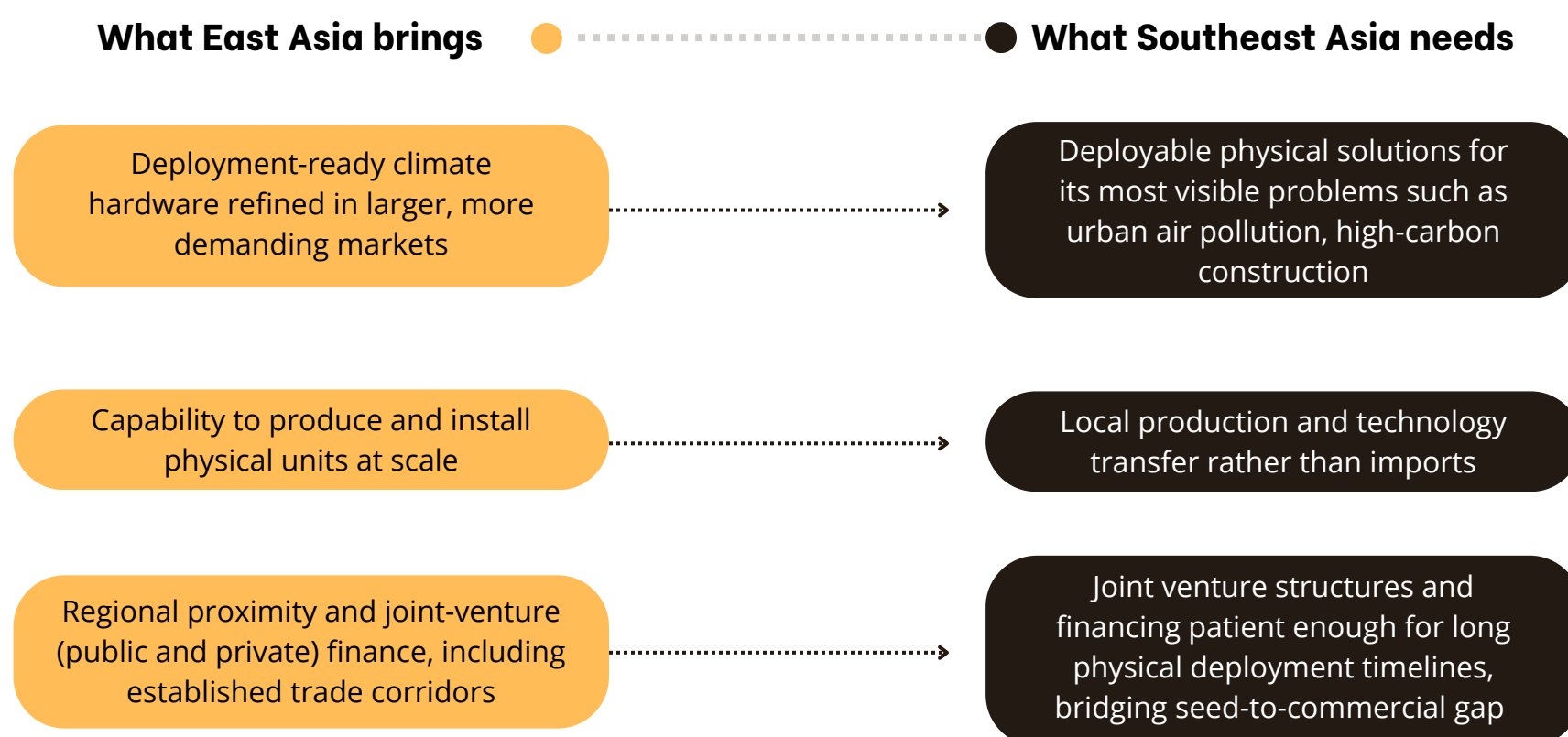


Figure 4: East Asia capability mapping against Southeast Asia's needs

Implications for East Asian entrants

If Europe's advantage is certification and compliance, East Asia's is production and proximity. Proven production at scale answers the region's most immediate and potentially profitable problems, and East Asian manufacturing capability makes it possible to produce locally rather than import. The East Asian entrant's advantage is speed to ground, with technology that works, produced nearby, deployed into demand that is already apparent.

4 Southeast Asia Climate Tech Supply Base and Marketplace

Entry conditions for climate tech are generally open across Southeast Asia, with most markets permitting majority or full foreign ownership in manufacturing and services while governments actively enable climate investment. **The challenge lies in identifying where each part of the operation belongs: where to produce, and where to sell.** Producing carbon output means securing land, sourcing feedstock and navigating emerging carbon-market rules, which favours the country with the material. Selling it means reaching buyers who already exist, which favours the country with the market.

The comparison below illustrates the entry reality for each role.

Indonesia

- Second-largest source of net land-use carbon globally
- Largest residue and biomass volumes in SEA; massive algal diversity (325+ species)
- New carbon exchange: IDX Carbon opened international trading in 2025, but thin volumes and low prices
- Land tenure often fragmented
- Evolving regulations ([PR 110/2025](#)), central-government carbon governance still consolidating

Singapore

- Self-positioning as Asia-Pacific's carbon trading hub, with voluntary carbon markets at its core
- Highest concentration of carbon service providers in Southeast Asia (>150)
- Singapore Carbon Market Alliance provides access to high-quality credits under Article 6
- Largest private green investment in ASEAN-6 (US\$2.68B, 33% of the total)
- Strong public support: >90% of residents aware of climate change, ~80% accept added costs for a low-carbon transition

Implications for new entrants

This produces a two-country model. Indonesia is the supply base with the region's largest biomass volumes and carbon sink potential. Meanwhile, Singapore is the marketplace, as it aims to be Asia-Pacific's carbon trading hub and where credible demand already sits. Produce in Indonesia through partnership and route the output to buyers through Singapore's established market.

5 Southeast Asia Climate Tech Sector Risks

For projects in climate tech, specifically related to carbon management, the dominant risks are structural rather than technological. It will be in the implementation or execution, not in the method itself. Three main risks exist across the region: 1) the absence of trustworthy local partners, 2) the immaturity of regulation and carbon markets and 3) the financing gap before project revenue exists.

Lack of local partners

Foreign entrants need a trusted local partner to navigate land access, permits and business norms that are hard to manage from offices abroad. Feedstock and demand can exist while entry is still stuck due to the lack of a trustworthy local partner, making partnership the practical prerequisite for everything else.

Regulatory immaturity (in both directions)

Heavy regulation and the absence of it could be a risk in and of itself. Lack of rigid rules create ambiguity over what qualifies, what is permitted and how credits are recognised. Southeast Asia's carbon frameworks are emerging and shifting, so the regulatory target might move as projects develop.

Project finance before revenue

Carbon-removal revenue arrives only after biomass is stored and independently verified. Combined with the region's not-so-demanding regulations, early-stage capital market, this timing gap is a structural risk for capital-intensive physical projects, not a question of demand.

Implications for European entrants

Europe's compliance and MRV credibility directly answer the regulatory-ambiguity risk and its blended-finance access eases the upfront-capital gap. The unmet need is the local partner, which is the one thing European provenance cannot supply from a distance and the first priority on entry.

Implications for East Asian entrants

Proximity is East Asia's advantage: closer partners, familiar business norms and the existing country or regional partnerships directly de-risk both the partner and finance gaps. Regulatory immaturity remains the shared challenge, best met by entering markets with clearer rules first.

6 Southeast Asia Climate Tech Market Entry Recommendations

Establish the carbon production supply chain in Indonesia and set Singapore as the main market. Beyond its raw endowment, Indonesia is the second-largest source of net land-use carbon globally. Article 6 of Paris Agreement trading is now formally enabled, letting Indonesian credits count toward foreign buyers' climate targets. Singapore turns that supply into revenue, with its concentration of carbon service providers and bilateral agreements giving the output a credible route to international buyers. Sell internationally through Singapore, deploy products where need is visible and treat the local partner as the first hire.

Sub-segment

Engineered Carbon removal, capture and utilisation

Regional supply base

Indonesia

Regional marketplace

Singapore

Five concrete moves follow:

1 Secure a local partner

This is not a task deferred until after setup, since this is a precondition before expanding the market, which will be useful for land access, permits and navigation. In guarded markets, structure as a joint-venture where a local party holds required control.

2 Establish production in Indonesia through partnership

Build the supply base where the material is largest: the region's biggest residue volumes and carbon sink potential, as well as Article 6 trading now enabled. Enter through a local partner for land access and expect to work with rules that are still consolidating.

3 Route sales through Singapore, not thin local demand

Domestic Southeast Asia carbon prices are low, so the viable model produces in-region and sells in Singapore or internationally through the country to reach corporate net-zero buyers among European and East Asian firms with Southeast Asia supply chains.

4 Underwrite the upfront-capital gap deliberately

Because carbon revenue arrives only after storage and verification, prepare for a cost-first timeline: secure buyer commitments early and access blended or purpose-built finance.

5 Lead with regulatory credibility and clean sourcing

Turn compliance into the selling point regulators and partners would be convinced to buy: rigorous MRV, deforestation-free traceability, durable-removal certification. In a market whose biggest reputational risk is credit integrity, provable quality is the competitive edge.

Work with Kala Advisory

This spotlight is the free sector read. **Your entry strategy needs a sharper one.**
Kala Advisory produces two things for businesses moving into Southeast Asia:

a Custom market reports

A sector-by-sector entry playbook built around your target market and industry, with market sizing, the entrant angle, the regulatory and compliance reality, and concrete, costed recommendations on where and when to enter.

A commissioned report is tailored to your sector, your risk appetite and your timeline.

b Advisory and entry support

Direct guidance on establishing in Southeast Asia: entity structuring, regulatory and tax navigation, partner and counterparty diligence, and on-the-ground market entry, drawing on a regional network of in-country analysts.

If this issue raised a question specific to your business, that is exactly where an engagement begins.

Website: kala-advisory.com • **Email:** business@kala-advisory.com • **Our media service:** sea-daily.com

Methodology and sources

Scope: A sector spotlight on Southeast Asian climate technology for European and East Asian entrants. Market sizing uses private green investment and nature-based carbon offset supply data; ownership, land and carbon-market rules are summarised at a directional level and are orientation, not legal advice. Figures blend the latest available vintages and are current to 2026.

Method: Three lenses are combined: sector momentum, entrant fit and entry feasibility; to identify a base market and a scale market. Risk assessment is qualitative, weighting the three risks most specific to a capital-intensive carbon sector.

Source index

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Southeast Asia Daily

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The Fight for Clean Air Has Many Fronts. One of Them Is Algae.

Posted on August 5, 2026 by Protasius Isyudanto



The device on Rehaan Lulla's desk looks like a lava lamp. A tall glass column, a slow swirl of green and a soft glow. What it is invisibly doing is drawing carbon dioxide out of the room and releasing oxygen more efficiently than the pot plants most people keep for the same reason. The organism inside is

microalgae, grown for decades as food and as a cosmetics ingredient, which Rehaan's start-up AlGreen has repurposed to clean the air.

The reason behind the invention is air pollution, which is not a niche or a distant concern. It is now ranked among the leading causes of death worldwide. Southeast Asia, the region AlGreen is most interested in, lives with it more acutely than most. Seen that way, the green column is not a room decoration but one of the solutions to something very large.

An Invisible Killer



Haze over Singaporean skyline, partly attributed to the forest fires in neighbouring Indonesian island of Sumatra. (Wikimedia)

Air pollution has quietly become one of the largest threats to human health. In 2021, it was estimated to be linked to 8.1 million deaths worldwide, making it the second-leading risk factor for death after high blood pressure.

It doesn't only pose health risks in the outdoors. Around 40% of the world's population still cooks and heats with wood, charcoal, dung or coal and the resulting household air pollution kills millions more each year. Because people spend most of their lives inside, either in homes, offices, classrooms or shops, the quality of indoor air matters as much as the haze outside.

The responses to all this vary and no single one is sufficient. Governments phase out coal and tighten emissions rules. Cities push public transport and electric vehicles. Households buy mechanical air purifiers that trap particles behind filters. Each helps but they have their own limits. Filters catch particles but do nothing for carbon dioxide, while the pot plants people keep for the same reason work slowly. It is in that gap that a certain biological approach starts to look interesting, one that led to the conception of Rehaan's initiative.

The Case for Spirulina



A spirulina culture. (AlGreen)

AlGreen began in a classroom. Rehaan was studying at the Hong Kong University of Science and Technology when he realised that biology, so

central to medicine and food, was barely being applied to the environment. He and a few classmates dug deep, looking for something that is of value and they could actually build. They ended up in an organism most people only knew as a health supplement. That organism is spirulina, a spiral-shaped microalgae long grown for its protein and sold in tablets and face creams. When the project started, Rehaan noted that there were fewer than ten companies anywhere working on microalgae for carbon removal.

He calls it tiny but mighty and the reason is simple enough. A leafy plant pours most of its energy into roots, stems and bark, while spirulina constructs almost none of that, so nearly all of its biomass turns carbon dioxide into oxygen. It photosynthesises across its whole tiny body, far faster for its size than a pot plant can. By one estimate, a single gram of it can absorb close to two grams of carbon dioxide. It also grows in water that can rarely be used for other purposes, so it doesn't need to compete with farmland.

From the Lab to the Living Room



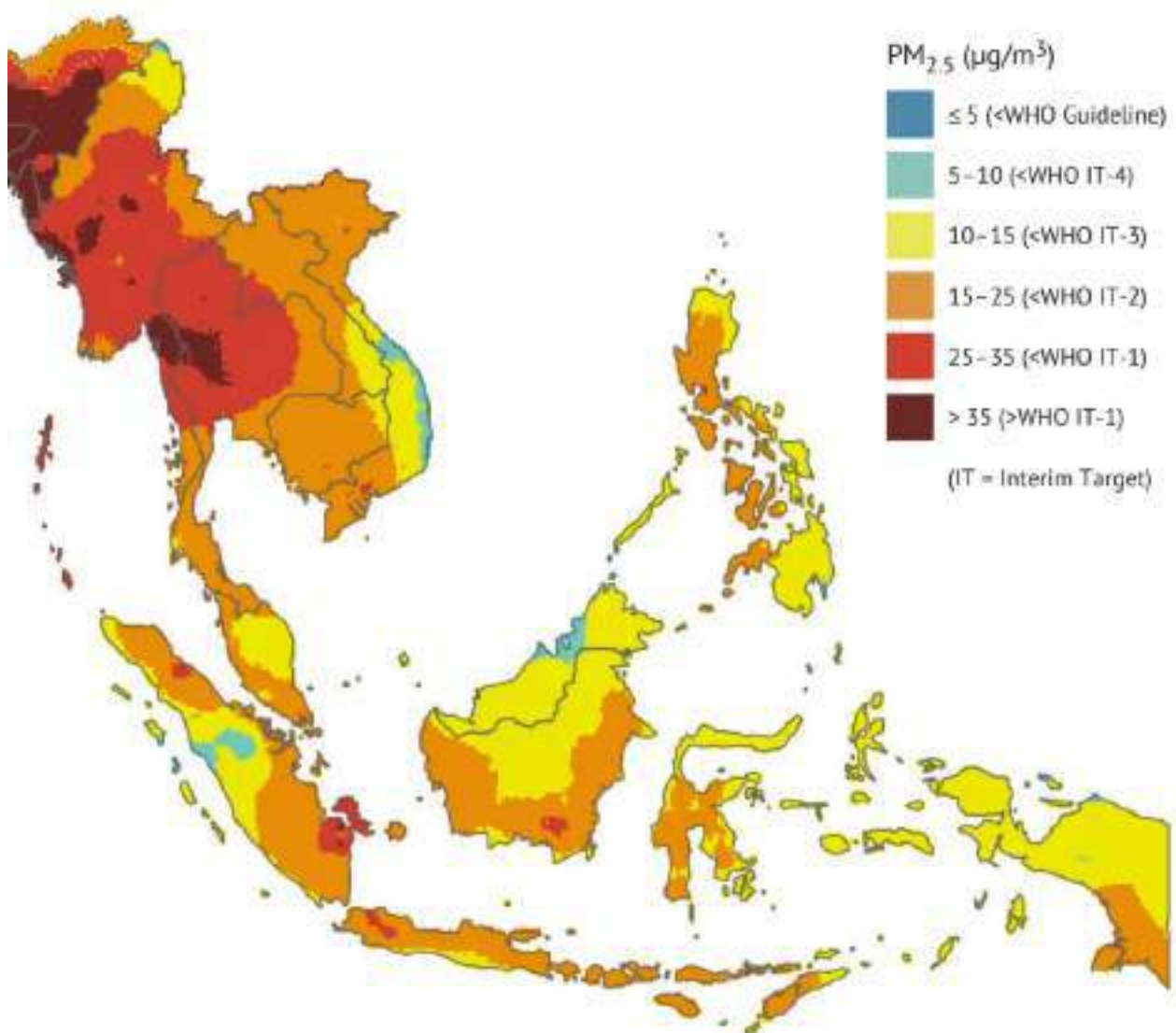
AlGreen's two main products: the table-sized VAYU and human-sized OAK. (AlGreen)

AlGreen's work has largely been to build the hardware that spirulina needs to do its job in ordinary places. In its case, the laboratory ends up being people's homes and offices. On one hand, VAYU is destined for those who keep their windows shut but still want cleaner air. On the other hand is the OAK, an adult human-sized microalgae tank which includes a carbon capture device and serves larger places like schools and universities.

Because clean air is invisible, the project's main challenge is proving the devices work. Rather than explaining the biology, AlGreen maintains that a single VAYU does the work of roughly 40 large house plants, a figure from its own tests backed by third-party testing. Meanwhile, its larger machines carry sensors that track how much carbon dioxide is absorbed and how much is produced, providing real-time performance data.

There is, however, an honest limit worth naming. AlGreen's devices capture carbon by drawing it into algae biomass, but capture only becomes removal if that carbon is durably stored rather than allowed to escape. For the roughly 700 VAYU users, the company's official guideline is to pour the spent algae down the sink or use it in the garden, where the captured carbon may return to the air. Nevertheless, its larger devices tell a different story. For the OAK tanks, AlGreen harvests the algae and converts it into biochar (which does not decompose and so locks the carbon away) or into storable bio-oil which Rehaan hopes one day to refine into aviation fuel.

Where the Air Is Worst and the Algae Grows Best



Annual average ambient PM_{2.5} exposure in 2019 in Southeast Asia, which has since evolved with Indonesia declared most polluted in 2025. (Health Effects Institute)

While air pollution is a global problem, Southeast Asia carries more than its share. In 2025, the most polluted country in the region, Indonesia, has an average PM_{2.5} level six times over the limit set by the World Health Organisation (WHO). To note, PM_{2.5} is potentially harmful airborne particles, which are small enough to slip past the body's defences and remain deep in the lungs. In 2024, the worst-ranked city in Southeast Asia was South Tangerang, on the edge of the country's capital Jakarta, where the annual average is roughly 12 times WHO's guideline. One study attributed more than 10,000 deaths a year in Jakarta alone to polluted air, alongside an economic cost that reaches billions of dollars.

The causes are partly seasonal and partly structural. Each dry season, land-clearing fires on Indonesia's islands of Sumatra and Kalimantan send haze across borders into Singapore and Malaysia, turning a national problem into a regional one, while coal plants around Jakarta and dense traffic keep the pollution reading high year-round. For such a region, any credible tool should be considered and here, AlGreen's scheme has an additional advantage. Spirulina grows fastest in warm climates, so the closer it is to the equator, the faster it grows. Where cold winters can stall the organism's growth in parts of East Asia, the tropical conditions across much of Southeast Asia are ideal for the species.

Rehaan lists India, Thailand, Malaysia and Indonesia as places where interest is building as the region already grows spirulina at scale for food and cosmetics. What's new is the idea of mobilising it against air pollution in a place whose climate is very suitable to Spirulina's growth.

The Green-Water Problem



AlGreen leads a learning session on spirulina and its applications. Education is a key part of developing the potential of this microalga. (AlGreen)

The obstacles to a concept like this are rarely the technology itself, but rather the perception of its potential users. From the outside, AlGreen's systems are a tank of green water, which is perceived as dirty water across much of the region, the kind found in a polluted river. It is, in fact, the other way around. Since the green is nutrient-dense water in which algae are thriving, the very thing that makes the device seem unhygienic is the sign the system is working.

A launch in the region could not simply be putting the product on a shelf. AlGreen would first have to explain what the machine is doing and actively shift the stigma. It is a branding and education cost rather than a flaw and it sits on top of a wider knowledge gap, since microalgae in the region is usually synonymous to nutrition and cosmetics, rather than clean air. A buyer has to be walked across two unfamiliar ideas at once, that algae can clean the air and that green does not mean dirty.

AlGreen's answer for that problem is to teach from the ground up. The company already runs workshops in schools, showing students why microalgae outperforms ordinary plants at absorbing carbon dioxide through simple experiments. The same approach is presumably how it would tackle the green-water stigma in a new market like Southeast Asia.

Spirulina is not the whole answer to Southeast Asia's air pollution problem and Rehaan would be the first to say there is no single answer. But the fit is real. In a region where the air is among the world's worst and the climate is among the ideal to the organism, there is real reason for optimism. Once the hurdle of perception is cleared, Rehaan's larger aim is to bring the living, breathing air purifier in the cities that need it most.



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Elisabeth Xavieria 📅 September 6, 2026

Ahead of the Indonesia-Eurasian Economic Union Free Trade Agreement (I-EAEU FTA) implementation scheduled for 2027, Indonesian President Prabowo Subianto proposed that the two biggest nations of the bloc, Indonesia and Russia, double their trade volume.



BUSINESS · INDONESIA

Indonesia Targets Mass Production of National Electric Vehicles by 2028

Elisabeth Xavieria 📅 August 15, 2026

President Prabowo Subianto said that Indonesia aims to kickstart the mass production of domestic electric vehicles (EVs) by no later than 2028. This ambition was announced during the inauguration of the National Electric Motorcycle (Molinas) in Bekasi, West Java province, on August 13. The president noted that “a large complex is being developed” across adjacent areas.



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Why Cooling the Planet Means Removing Carbon, Not Just Emitting Less

Posted on July 31, 2026 by Protasius Isyudanto



Excess carbon in the atmosphere is one of the most pressing problems of our time and Johanna Bröll's answer to it starts with something most people would set on fire. Every year, vast amounts of waste wood are produced worldwide and most of it is

either burned or left to rot, releasing carbon that a tree pulled from the air locked into its wood over decades. Rather than let that waste wood burn, Johanna's Berlin-based start-up, Carbonsate, buries it instead, sealing it underground where the carbon stays locked in the wood for centuries, removed from the cycle and turned into carbon credits.

Johanna has always been drawn to big challenges and to the simple solutions that are often underestimated. When it comes to the question of climate change, her answer is putting to use what is already around her. "I noticed that there are large amounts of waste wood all around the world and many of the waste wood streams are still unused," she says. "Instead of burning the waste wood, how could we preserve the carbon that is already embodied in the biomass to remove it from the cycle?" She landed on a simple method called biomass storage, in which they let plants do the hard work of capturing carbon dioxide from the atmosphere, then seal the wood in engineered underground chambers so the carbon cannot return to the atmosphere. Other carbon removal methods exist, but most are costly, complex and hard to scale. Biomass storage is a carbon removal method that is simpler and more durable, because it relies on a process nature already runs.

"Carbon Removal is hotter than Paris today"



A pharmacy sign in Paris showing a temperature of 44°C on June 26, 2026, far exceeding the the average June peak of 23.4°C. Heatwaves across Europe have become more prevalent and existing infrastructures experience difficulties to adapt. (Wikimedia/FreCha)

That was the slogan on Carbonsate's stand at the VivaTech event in Paris last June—one of the largest tech exhibitions in the country. It was, indeed, a hot day in the middle of a Europe-wide heatwave, which only made the slogan ring truer. Climate change is no longer an assumption or a talking point. It has reached people's daily lives and keeping the rise in the earth's temperature below 1.5°C—a commitment made by hundreds of countries under the landmark 2015 Paris Climate Agreement—now demands efforts both big and small.

Global warming, the driver of climate change, occurs due to carbon dioxide accumulating in the atmosphere which works like a blanket around the earth, trapping heat. Plants like trees work as natural carbon sinks, pulling carbon out of the air, absorbing it into their organic mass and storing it in their wood as they grow. But that storage is fragile, since when a tree is burned, the carbon goes straight back into the atmosphere. This is where a crucial

distinction lies. Offsetting means that for every tonne of carbon emitted, an equivalent amount is avoided or reduced somewhere else. Meanwhile, carbon removal means physically taking carbon out of the atmosphere and locking it away, whether by drawing it into soils, capturing it directly from the air with machines, or in Carbonsate's case, burying the wood that already holds it in sealed underground chambers.

From Invasive Bush to Carbon Credits

Carbonsate team members inspecting woody biomass before permanent underground storage. Only suitable, low-value biomass that would otherwise be burned or left to decay is used. (Carbonsate/Stefan Redecker)

So far, Carbonsate has operationalised one project in Namibia, with plans to implement the scheme in Bulgaria, Colombia and Cameroon. When implementing the biomass storage in a project, Johanna had to make sure the feedstock is suitable. First, the waste has to be woody, meaning that it is high in lignin, low in nutrients and low-quality wood that would otherwise be burned, never valuable timber. These are necessary to ensure that the wood waste does not decompose. Second, it must come from nearby the burial site,

since transporting biomass over long distances would generate the very emissions the framework tries to remove. Lastly, no waste wood should come from deforestation or anything that destroys an ecosystem.

The Namibian initiative shows what those qualifications look like on the ground. There, invasive bush spreads across large farms, a problem known as bush encroachment and Carbonsate identified it as the right source of biomass. The team checked for permits, made agreements with the landowners and harvested the bush, leaving it to sit for a few weeks, so it dried and the leaves fell off. The biomass was chipped and transported a short distance to the storage site, where the wood chips were measured for moisture and weight, to know how much carbon the material holds. The bush was then stored in an engineered underground chamber, around four metres deep, and sealed.

Although biomass storage is one of the carbon removal methods deemed simple, the main challenge is proving that it works and that the carbon stays locked away for centuries, removed from the air. Since the waste wood already holds the carbon, the job is to stop it escaping, so the chamber is conditioned to stay low in oxygen and low in moisture, where the wood cannot decompose. The storage site comes with sensors which measure five things: moisture and oxygen, which must stay low; methane and carbon dioxide, which would signal decomposition happening when it should not; and temperature. To prove the carbon is locked away, Johanna does not rely on the sensors alone. An auditor comes to the site, and a third party verifies the measurements, the calculations and the sustainability data. “It’s not just us claiming it,” she says, “but there are also externals who verify all that.” Only once it is signed and certified is a carbon credit issued for each tonne of carbon dioxide stored, ready to be sold and delivered.

Carbonsate works with two models. In the first, the start-up purchases the waste wood from its owners and implements the scheme itself. In the second,

landowners implement their own projects and keep the entire carbon credit revenue, while Carbonsate acts as a technology and service provider, supporting them with the method and the knowledge, while helping them to sell the carbon credits. The costs before revenue are covered by Carbonsate, by the project owner, or through dedicated project finance. For the latter, Johanna typically approaches local banks, which have a keen interest in supporting the region's economy or international capital providers aiming for sustainability investments. Once the carbon credits are issued, buyers from all over the world can purchase them.

Every carbon credit is a tradable certificate representing one tonne of carbon dioxide kept out of the atmosphere or removed from it. Companies buy them to either compensate for emissions they cannot yet eliminate themselves or as part of their own voluntary climate contribution strategy, which naturally supports high quality carbon removal projects. But not all credits are the same. Some come from avoidance initiatives that prevent emissions (such as protecting a forest) while others come from removal projects like Carbonsate's, which take carbon already in the air and lock it away.

These carbon credits themselves can be sold anywhere in the world, as this is a completely global market. Johanna mentions that her typical buyers are technology companies or firms with net zero commitments or carbon reduction targets validated by Science-based Target Initiatives (SBTi), mostly in Europe and the US. The market still has to grow, she notes, as much of the carbon market is still voluntary rather than mandatory, although she believes this will gradually change and climate leaders are now looking for more durable carbon removal schemes. Aside from these two regions, Johanna adds how she sees interest emerging in places including Southeast Asia, particularly Singapore.

"A Region with Enormous Potential"

Acacia nilotica in Indonesia's Baluran National Park, an invasive plant species that has grown in an area of 6,000 hectares in the natural reserve. These kinds of biomass can be used in Carbonsate's scheme.

(Ekuatorial)

Southeast Asia is one of the world's most vulnerable regions to the effects of climate change, with rising sea levels eroding coastal communities, as well as storms and typhoons bringing heavy rainfall and severe flooding that damage homes and livelihoods. Efforts to reduce carbon—the driver of climate change—have begun across the region and Southeast Asia does not shy away from contributing to global climate action. At the COP30 United Nations (UN) Climate Change Conference in Brazil in November 2025, the region's bloc, ASEAN, hosted its first-ever pavilion, showcasing regional initiatives and forging carbon trading partnerships, alongside collective efforts to advance energy transition, climate finance, nature-based solutions and a just transition. Among its numerous climate efforts, nearly all of it is aimed at emitting less, rather than removing what is already in the air. Carbon

removal, through methods like biomass storage, remains the region's untapped potential.

The raw material is certainly there. Southeast Asia's agriculture and forestry sectors generate more than 500 million tonnes of biomass residues a year, much of it low-quality wood that would otherwise be burned or left to decay, which is exactly what Carbonsate looks for. Asked whether the region's material is as workable as Namibia's or Cameroon's, Johanna does not hesitate to say yes. "I feel that the region has enormous potential, simply because the availability of woody biomass is a given there." For an initiative aiming to remove more than 10 million tonnes of carbon dioxide from the atmosphere every year by 2033, a region setting fire to its feedstock is hard to ignore for Carbonsate.

When asked what stands in the way, Johanna does not point to the method but to what she calls an ecosystem of carbon removal. The technology, she is clear, is rarely the obstacle. Biomass storage is new enough that many people have never heard of it and the reaction she often meets is that it sounds too good to be true, so much of her work becomes convincing people that something this simple genuinely works. Then comes regulation, where supportive rules would help and, just as importantly, the absence of any rules creates ambiguity about what is even allowed. But the harder test is execution. Adapting to unfamiliar regulations, cultures, and ways of doing business, market by market, is where the real difficulty lies.

The Burning Question

Burning releases the carbon stored in woody biomass back into the atmosphere. Carbonsate's approach preserves suitable woody biomass underground instead, creating durable carbon removal.

(Carbonsate/Stefan Redecker)

In much of Southeast Asia, waste wood is not treated as waste at all, but as fuel. Across the region, agricultural and forestry residues are burned to generate power and ASEAN's own energy strategy frames biomass as a renewable, low-carbon alternative to fossil fuels, on the logic that the carbon released is reabsorbed by new plant growth. The region still leans heavily on coal, which generated around half its electricity in 2024, so cleaner-burning biomass is widely cast as part of the answer rather than the problem.

This leaves a burning question. In a region that depends on burning this wood for power, why bury it instead? Johanna doesn't believe every piece of biomass should be burned simply because it can. Burning wood, including as fuel, releases all its carbon at once and as a molecule that carbon ends up in the atmosphere in the end. If stored permanently instead, the same wood becomes a carbon resource that generates durable credits and as Carbonsate's projects show, waste wood could even become a new export

product or a fresh revenue stream for rural communities. There is a real case for burning biomass. The method is cheap, local and renewable enough. But Johanna’s argument is hard to dismiss. Energy can be found in many places, while carbon—once released—is almost impossible to call back.



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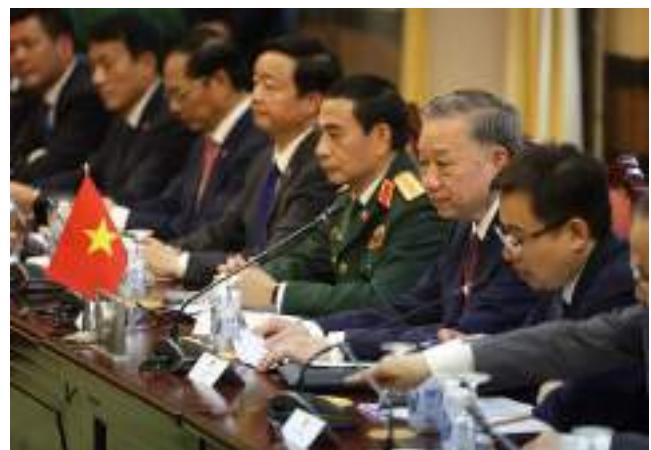


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Our Commitment to **Sustainable Travel**

Throughout Nabguide's tour operations, we aim to plan every journey so that it does no harm to people and the planet. It was both an honour and a pleasure to host the delegation from CASE for South East Asia, a group whose values align closely with how we operate when welcoming guests, whether for leisure or business.

Across every service delivered, we made deliberate efforts to keep our practices sustainable, from accommodation, transportation, meals and other logistics. Prior to departure, printed travel guidebooks were replaced with an interactive digital version featuring smart links, reducing paper use while improving accessibility. The printing budget was reallocated to a pre-departure briefing at Sate Khas Senayan, CGK airport, where participants gathered over light refreshments to review the program and receive final travel information.

The following outlines the steps taken in each area.

1. Accommodation

The delegation was hosted at **OKKO Hotels Paris** Porte de Versailles, part of OKKO Hotels, the first hotel group in France to adopt société à mission status, registered in 2023. A société à mission is a legal status created under the 2019 PACTE law: a company writes its environmental and social commitments directly into its corporate statutes and is legally bound to act on them, rather than simply publishing a voluntary charter. Okko's stated purpose centres on combining high-end hospitality with ecological sobriety, while respecting its social environment.

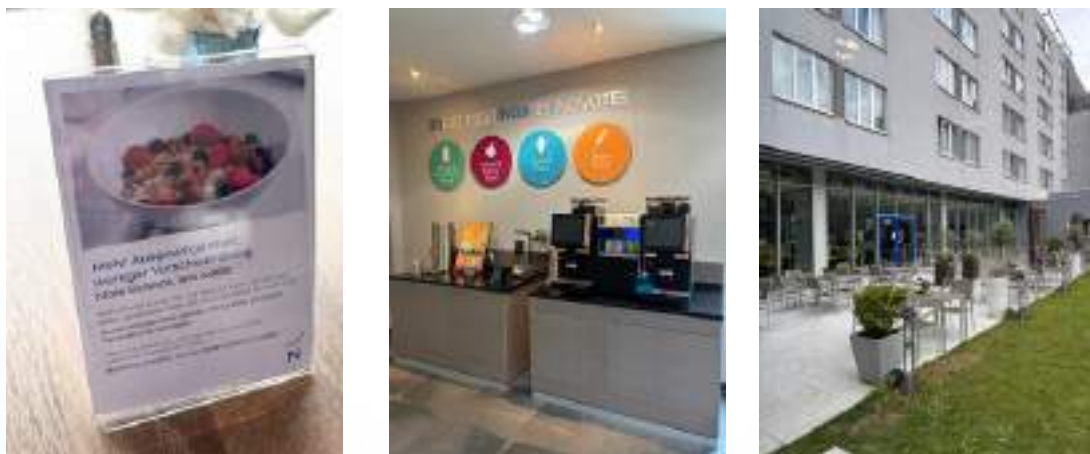
These commitments are visible throughout the property's everyday operations. Guests can decline daily housekeeping to save resources, with a donation made to a partner association for each declined service. Plastic bottles are replaced by micro-filtered water fountains on every floor, and room cosmetics are made from 90 to 97% natural ingredients in eco-designed packaging. The bedding uses natural-fibre linen developed to reduce both carbon footprint and water consumption, and it requires no ironing. Energy use is kept to what is needed, with shared facilities such as the sauna heated only on demand rather than continuously. Waste is sorted into separated recycling bins in the rooms and across the hotel's shared spaces, and the bar and lounge operate on a no-waste basis, offering drinks and snacks without single-use excess.

Our Commitment to **Sustainable Travel**

Selecting this property allowed Nabguide to align the delegation's stay with an accommodation partner whose sustainability commitments are formalised and reflected in its day-to-day operations.



In Cologne, the delegation stayed at **Novotel Köln City**, a hotel certified under Green Key, an international sustainability label managed by the Foundation for Environmental Education. Green Key certification requires a hotel to meet recognised criteria across environmental management, water, energy, and waste, as well as staff and guest engagement, verified through an on-site audit and renewed periodically. These commitments are visible in the hotel's daily operations: dining areas display reminders encouraging guests to take only what they will eat to reduce food waste, and the meeting spaces offer a structured food concept balancing healthier and functional options. Some of these practices are shown in the photographs below.



Across both cities, Nabguide selected hotels whose sustainability commitments are not only certified but embedded in daily operations, evident in how they manage water, energy, waste, and guest services. These choices reflect how Nabguide puts the values of sustainable tourism into practice, delivering a high standard of service to the delegation without compromising on environmental responsibility.

Our Commitment to **Sustainable Travel**

2. Transportation

Nabguide's first priority in any programme is to ensure that delegates can carry out their activities as efficiently as possible. Given the density of the IESR delegation's agenda, with multiple institutional meetings across several venues each day, a dedicated coach was the most practical and responsible option. A single 37-seater bus, supported by a van, allowed all fifteen delegates to move together directly between venues, keeping to a demanding schedule while accommodating split departures and last-minute adjustments. Compared with dispatching multiple individual vans, consolidating the group into one vehicle reduced both the number of journeys and the associated carbon emissions, while public transport alone would not have met the timing and coordination the programme required.

Beyond the main programme transport, we also made available a more sustainable option for leisure travel by arranging discounted access to Turtle, a Paris-based electric bike-taxi service. Turtle's electrically-assisted bike-cabs are around 95% less polluting than a conventional car and can use bike and bus lanes, offering delegates a low-impact way to discover the city's monuments. Providing this option reflects Nabguide's standard practice of offering environmentally responsible alternatives to guests wherever possible.



Our Commitment to **Sustainable Travel**

3. Catering and meal

Across the programme, Nabguide coordinated meals to support both participant welfare and the wider goal of minimising waste. Menus were planned around the delegation's tastes and dietary requirements, with halal needs and allergies confirmed in advance and quantities coordinated with each vendor to match the actual number of participants. Following the catering arrangements for the Paris office meetings, meal orders were adjusted to reflect actual attendance, which was lower than the original proposal assumption of 23 participants across 5 catering sessions. Rather than ordering excess food, the unused budget was transparently reallocated—with prior coordination and approval from the EO team—to provide an additional group dinner for delegates arriving late in Cologne after a long travel day. This approach reflected Nabguide's commitment to flexibility, transparency, participant welfare, and responsible resource management.

Reducing food waste was a consistent priority. Portions were planned to be eaten in full, and where food remained, delegates were able to take leftovers in hygienic boxes to share with colleagues rather than discard them. Menu choices also kept emissions in mind: while some meals included higher-emission dishes such as beef, lower-emission options including chicken, seafood, and vegetarian or vegan dishes were consistently available, helping to reduce the overall carbon footprint of catering. Some of these meals are shown in the photographs below.



Conclusion

Nabguide's commitment to sustainable travel runs through every part of the services delivered for this programme. At each stage, the aim was to meet the delegation's needs fully while avoiding unnecessary harm to people and the planet. By working with local businesses and coordinating closely with the delegation, Nabguide was able to find practical common ground, delivering a high-quality experience while keeping choices as environmentally responsible and as healthy as possible.

This approach was consistent across accommodation, catering, and transport, where Nabguide favoured lower-emission and lower-impact options wherever they could be reconciled with the delegation's requirements. Participants were also provided with a reusable tote bag, serving both as a practical, presentable way to carry materials and as a small step toward reducing single-use plastic. Nabguide was delighted to support the CASE for South East Asia delegation on this programme and remains committed to delivering services that are reliable, responsive, and as sustainable as possible.

CLIMATE TRANSITION IN BRAZIL: SOCIAL IMPLICATIONS OF ADAPTATION POLICIES

POLICY BRIEF

Amélie CLEMOT, Katya DIAZ SALCEDO, Lidwina PUTRI, Thomas SCHOBER

OVERVIEW

SciencesPo Master in Advanced Global Studies students in partnership with UNESCO analyzed Brazil's climate adaptation policies and its impacts on vulnerable groups.

Brazil is actively advancing its sustainable development agenda through the implementation of different climate change policies, which prioritize the identification of vulnerable groups and recognize their heightened susceptibility to climate impacts. However, the implementation of these policies often falls short, risking maladaptation that worsens inequalities.



KEY SECTORS AND CASE STUDIES



1. AGRICULTURE

Sectoral Overview:

Agriculture is crucial to Brazil's economy, accounting for a significant portion of GDP and employment. Despite its contributions, agriculture also represents a major source of GHG emissions. Climate-smart agriculture (CSA) practices have been identified within a range of sustainable solutions, increasing resilience and productivity while reducing emissions.

Challenges: Limited access to credit and technical assistance for small-scale farmers. Disparities in the adoption of climate-smart agriculture practices.

Consequences for Vulnerable Groups:

- Smallholders, particularly women and black Brazilians already suffer from lower productivity, less access to subsidized credit, technical assistance and markets.
- This can further exacerbate inequalities and hinders smallholders' ability to catch-up and improve their livelihoods. This risks the unnecessary persistence of poverty and limits Brazil's environmental potential.

RECOMMENDATIONS:

- **Public support programs:** Tailor public support programs including access to credit and technical assistance towards small-time farmers.
- **Micro targeting communication:** Target local farming cooperatives, TV and radio stations to spread the word about enhanced public support programs for small-time farmers so that take-up rates for credit for climate smart agriculture initiatives increase.

2. WATER-ENERGY

Sectoral Overview:

Over 60% of Brazil's electrical energy comes from hydroelectric power. While renewable, hydroelectric plants have environmental and social impacts, including land flooding and displacement of local populations. Efforts are underway to balance energy needs with socio-environmental concerns.

Challenges:

The expansion of hydroelectric power has led to environmental degradation and the displacement of communities.



Consequences for Vulnerable Groups:

- Rural communities, including farmers and indigenous populations, face loss of land and livelihoods due to the construction of hydroelectric plants.
- The relocation often results in diminished access to natural resources, cultural dislocation, and economic instability, making it harder for these groups to sustain their way of life.

RECOMMENDATIONS:

- **Integration of Comprehensive Corporate Social Responsibility Policies:** Ensure community participation throughout the hydroelectric project including in the planning, implementation and operation stages. This participation can help ensure that affected populations have access to a dignified and sustainable life.
- **Strengthening State Agencies:** Enhance the capacities of state entities responsible for monitoring the social impacts of private investment projects, ensuring robust oversight and accountability.

- **Building and Strengthening Civil Society:** Encourage and reinforce the development of civil society organizations affected or potentially affected by hydroelectric projects. A strong civil society can advocate for and ensure that rights are upheld throughout all project phases and is important for the identification and support of vulnerable subpopulations requiring special attention.
- **Establishing Compensation Policies:** Implement compensation policies for populations that have been negatively affected by hydroelectric projects, ensuring redress and support for these communities.

3. HEALTH - WATER

Sectoral Overview:

To mitigate the health effects of climate change, enhance resilience and improve water quality, Brazil implemented the First Water Cisterns programme. This programme harvests rainwater through cisterns and sterilizes it for consumption. The programme has had a positive impact on infant health.

Challenges:

Climate change impacts public health through intensified disease transmission, water scarcity, and extreme weather events.



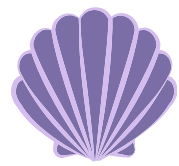
Consequences for Vulnerable Groups:

- The poorest and least educated women had difficulty adhering to the water treatment training.
- Vulnerabilities are magnified for children, the elderly, and pregnant women, leading to heightened risks of malnutrition, waterborne diseases, and heat-related illnesses.

RECOMMENDATIONS:

- **Improve Access to Health Systems for Wider Communities:** More accessible health systems specified to the needs of the vulnerable groups. Health care can be tailored for the specific needs of indigenous communities
- **Implement risk communication and community engagement for risks related to public health emergencies due to climate change:** Vulnerable communities should be engaged and empowered to take on leadership roles when informing the risk.

- **Integrate education on climate change and the impacts on health in school curriculum:** Integrate climate change education and the effects on health in interdisciplinary school subject, such as through discussion on local environmental issues such as air pollution and its impacts towards health.
- **Adapt the materials related to the implementation of policies so they are accessible for people of all educational levels:** Simplify the training for people who benefit from the policy implementation such as the cisterns program so it is more comprehensible for people with different educational levels



4. COASTAL MANAGEMENT

Sectoral Overview:

Brazil's extensive coastline faces challenges from diverging land use interests, environmental degradation and climate change. The Orla Project aims to protect coastal areas through integrated management, emphasizing local participation and sustainable use of resources.

Challenges:

Lack of political will, difficulties with bureaucracy and coordination, gaps in funding and in training particularly for public officials and proactive solutions are important roadblocks to coastal management and the implementation of coastal adaptation solutions.

Consequences for Vulnerable Groups:

- Artisanal fishermen, unemployed youth, and traditional communities residing in coastal areas face increased risks due to environmental changes and poor policy implementation.
- Lack of effective management and participation in implementation stages can result in reduced resources, loss of livelihoods, and exacerbated social inequalities.

RECOMMENDATIONS:

- **Enhanced Coordination and Governance:** Establish a dedicated coastal management authority to ensure better coordination between federal, state, and local governments.
- **Community Involvement and Educational Programs:** Strengthen the role of local communities in the planning and decision-making processes by enhancing inclusive participatory mechanisms. Implement educational programs for locals and tourists that teach coastal best practices.
- **Legal Framework:** Create legal instruments that institutionalize coastal management plans and provide the necessary legal backing for their enforcement, reducing dependency on the political will of changing administrations.

- **Funding and Resource Allocation:** Prioritize budget allocations for the maintenance and implementation of the Integrated Management Plans (PGIs) at the municipal level, ensuring that funds are earmarked specifically for environmental conservation and sustainable development projects.
- **Monitoring and Evaluation:** Establish robust mechanisms for the continuous monitoring and evaluation of coastal management policies such as clear metrics for assessing the effectiveness of policies and adapting them based on environmental changes and emerging scientific data.



CONCLUSION

- Addressing the needs of vulnerable communities and overcoming implementation challenges are critical to achieving comprehensive and equitable climate resilience.
- Strengthening policies, enhancing participation, and prioritizing local needs will be essential for Brazil's continued progress in climate adaptation and mitigation.
- It is crucial to incorporate an environmental justice approach, which should emphasize recognition, distributive, and participatory justice for those most affected by environmental issues
- Integrating a gender-sensitive perspective into all training and skills development initiatives related to the climate transition is necessary, with a particular focus on supporting vulnerable groups.